

Holonomy descent and generator control

Split-Zero research programme

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This companion retains the original holonomy source and coefficient frames, the relation norm, the complete complex-parameter comparison and the resulting control of the original arithmetic generator. The source note is reproduced in full; its H49c inverse domain is made precise in HDI4–8. The completed-quotient calculations in CQ, HDI and HCD are retained as independent proofs of the same maps, with the continuous cochain construction and weighted relation cross term spelled out in HCD. The final CR proof gives the continuous-source completion and degree limit at the original fixed packet.

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All-holonomy descent of the original theta quotient

Result and scope

The original arithmetic source is recovered exactly by the complete family of twisted circle periodizations. The new calculation retains what happens when minimization over the original theta relations is performed in that family. There is a positive, explicitly calculated difference between the original quotient metric and the average of the fibre quotient metrics. It is the norm of a family of original relations, not an unspecified error.

If every phase-source Gram lies between $(1 - \eta)M$ and $(1 + \eta)M$, where M is their exact mean and $0 \leq \eta < 1$, the calculated comparison is

$$(1 - \eta^2)G \preceq \overline{G} \preceq G, \quad G - \overline{G} = \int (C - C_\theta)^* M_\theta (C - C_\theta) \frac{d\theta}{2\pi}. \quad (\text{H1})$$

Here C and C_θ are minimum-norm sections of the **same** original polynomial quotient. Every difference $C - C_\theta$ has its actual theta primitive. The factor η^2 is sharp for the stated matrix hypotheses. This is an application of positive-matrix variational methods, not a priority claim for operator concavity.

The family is joined to finite cyclic covers, the new residue-constituent coefficient, and the original endpoint-volume criterion. All these constructions remain over the marked split base. No finite-field weight theorem is applied to a Hilbert circle quotient. No uniform arithmetic upper bound contradicting the inherited quartet threshold is proved in this note.

1. Fixed inputs, maps, and the marked split base

The owner's scalar object, its support observation, and its arithmetic observation are retained:

$$G(R) = \{\tau\} \sqcup R^\bullet, \quad e_R = 0_R^\bullet, \quad p_R(\tau) = 0, \quad p_R(r^\bullet) = r, \quad b_R(\tau) = 0_{\mathbb{B}}, \quad b_R(r^\bullet) = 1_{\mathbb{B}}. \quad (\text{H2})$$

The map (p_R, b_R) is injective. The arithmetic coefficient square is

$$\begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p_{\mathbb{Z}} \downarrow & & \downarrow p_{\mathbb{C}} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array} \quad (\text{H3})$$

The target of $p_{\mathbb{Z}}$ is infinite. The pointed-monoid base has structure map $\{\tau_0, 1\} \rightarrow U_{\times} G(R)$, taking τ_0 to τ and 1 to $1^\bullet$. Direct image to its point retains the coefficient object; it is not replaced by its Boolean localization.

On a fixed original support label ℓ , a coefficient-linear map f lifts by

$$(\ell, v) \mapsto (\ell, fv), \quad \tau \mapsto \tau. \quad (\text{H4})$$

For a relation submodule B_ℓ , the quotient is $(\ell, v) \mapsto (\ell, [v])$. Its equality criterion is $v - v' \in B_\ell$. Thus $f(v) = 0$ gives the receiving fibre zero, not external absence. Every map constructed below has this lift. Products of phase fibres use the original product support diagram; on a common label the scalar Fourier matrices give linear equivalences. For unequal labels, sums first use the stated joins and transports. No injectivity of arbitrary support transports is assumed.

Retain the original theta data

$$\begin{aligned} V &= \{\phi \in \mathcal{S}(\mathbb{R})_{\text{ev}} : \phi(0) = 0, \int \phi = 0\}, \\ \Theta\phi(x) &= 2 \sum_{n \geq 1} \phi(nx), \quad Q = \mathcal{B}/\Theta V, \quad D = -x\partial_x, \quad g = 2\xi. \end{aligned} \quad (\text{H5})$$

The prior full-jet analytic construction supplies $F_h \in \mathcal{B}$ with $\mathcal{M}F_h = g/h$, for a selected finite packet h with every selected order complete. The present finite arithmetic module is

$$E = \mathbb{C}[S]/(\chi_{h,k}), \quad q = \deg \chi_{h,k} > 0, \quad A = M_S, \quad S = s_1 + \cdots + s_k. \quad (\text{H6})$$

Use its same source map and full inclusion:

$$\mathcal{V}P = P(D_1 + \cdots + D_k)F_h^{\otimes k}, \quad J^{(k)}\mathcal{V} = \eta\pi_\chi, \quad q^{(k)}\mathcal{V} = \sigma_h^{\otimes k}\eta\pi_\chi, \quad (\text{H7})$$

$$\eta[P] = v_h^{\otimes k}P(A_k)1, \quad v_h = j_h(g/h). \quad (\text{H8})$$

The unit in (H8), every nilpotent jet, and $g = 2\xi$ stay in these equations. For $N \geq q - 1$, let $P_N = \mathbb{C}[S]_{\leq N}$, $J_N = \pi_\chi|_{P_N}$, and $B_N = \ker J_N = \chi P_{N-q}$, where $P_{-1} = 0$. Fix their coefficient bases. A matrix denoted B below is the injective coefficient matrix for multiplication by this literal χ ; it is not a replacement of a higher contracted relation ideal by a power of χ .

The new source corrections are incorporated as follows. The supplied primary-source correction resolves the reference at Weil II 3.7.4 as Roman I.8.11 in Weil I. The strict bound in 3.2.10, the square in 3.2.13, and the lisse/mixed hypotheses in 1.8.12 remain. The supplied translation note keeps its scalar $\exp((- \Phi(-a) - ta)/u)$; the marked arithmetic fibre remains $a = 0$. The supplied residue note's exact rank-one and curvature identities are inherited inputs in Section 8, not discoveries of this continuation.

2. All circle holonomies recover the full source

Retain the predecessor's exact logarithmic coordinates

$$r = \log x_k, \quad z_i = \log x_i - \log x_k \ (i < k), \quad (\mathcal{U}_k F)(r, z) = e^{(kr + \sum z_i)/2} F(e^{r+z_1}, \dots, e^{r+z_{k-1}}, e^r). \quad (\text{H9})$$

Their logarithmic Jacobian is one. With $\mathcal{K} = L^2(\mathbb{R}^{k-1}, dz)$, this is an isometry into $L^2(\mathbb{R}, dr; \mathcal{K})$, and

$$\mathcal{U}_k D^{(k)} = (-\partial_r + k/2)\mathcal{U}_k. \quad (\text{H10})$$

No relative variable is integrated out of the vector-valued source. Integrals in this note are defined inside each specified coefficient fibre, then lifted by (H4); no unconstructed integration operation on arbitrary mixed-support carriers is assumed. For $L > 0$ and $0 \leq \theta < 2\pi$, define

$$(\mathcal{Z}_{L,\theta}\psi)(r) = \sum_{a \in \mathbb{Z}} e^{ia\theta} \psi(r + aL), \quad 0 \leq r < L. \quad (\text{H11})$$

The original finite sources have sufficient weighted Schwartz bounds to differentiate and sum this expression on compact sets. It extends to the Hilbert-space transform by density. Its boundary and Fourier equations are

$$\mathcal{Z}_{L,\theta}\psi(r + L) = e^{-i\theta} \mathcal{Z}_{L,\theta}\psi(r), \quad \widehat{\mathcal{Z}_{L,\theta}\psi}(n) = L^{-1/2} \widehat{\psi}((2\pi n + \theta)/L), \quad (\text{H12})$$

where $\widehat{\psi}(u) = \int \psi(r) e^{iur} dr$, and the target basis is $L^{-1/2} e^{-i(2\pi n + \theta)r/L}$. To check the factor, substitute $y = r + aL$ in the coefficient integral. The two phase factors give $e^{ia\theta} e^{-ia(2\pi n + \theta)} = 1$.

Exact isometry and inverse. The full transform is

$$\mathcal{Z}_L : L^2(\mathbb{R}; \mathcal{K}) \xrightarrow{\sim} \int_{[0, 2\pi)}^{\oplus} L^2([0, L]; \mathcal{K}) \frac{d\theta}{2\pi}. \quad (\text{H13})$$

For $0 \leq r < L$ and $a \in \mathbb{Z}$, its inverse is

$$\psi(r + aL) = \frac{1}{2\pi} \int_0^{2\pi} e^{-ia\theta} (\mathcal{Z}_L \psi)(r, \theta) d\theta. \quad (\text{H14})$$

For each r , these are precisely the Fourier-series map and inverse on the sequence $(\psi(r + aL))_a$. Parseval followed by integration in r proves (H13). The measure $d\theta/(2\pi)$ is the actual Fourier inversion measure; no arithmetic mass has been assigned the value one.

For the fixed source-column map $\Psi_N = \mathcal{U}_k \mathcal{V}|_{P_N}$ put

$$M_N(\theta) = (\mathcal{Z}_{L, \theta} \Psi_N)^* (\mathcal{Z}_{L, \theta} \Psi_N), \quad M_N = \Psi_N^* \Psi_N. \quad (\text{H15})$$

Then, exactly,

$$\int_0^{2\pi} M_N(\theta) \frac{d\theta}{2\pi} = M_N. \quad (\text{H16})$$

The original norm is $\int |P(k/2 + iu)|^2 m_{h,k}(u) du$, where $m_{h,k} = w_h^{*k}$ and $w_h = |(g/h)(1/2 + iu)|^2/(2\pi)$. Partial Plancherel therefore gives

$$M_N(\theta)_{ab} = \frac{2\pi}{L} \sum_{n \in \mathbb{Z}} m_{h,k}((2\pi n + \theta)/L) \overline{P_a(k/2 + i(2\pi n + \theta)/L)} P_b(k/2 + i(2\pi n + \theta)/L). \quad (\text{H17})$$

For $k \geq 2$ all these scalar weights are positive. Infinitely many distinct sample points then imply positive definiteness on every finite polynomial source. The uniform quantitative statements below instead use the displayed relative source bound, which by itself implies positive definiteness, including other admitted sources.

At $\theta = 0, n = 0$, retain the term $L^{-1} \widehat{\Psi}_N(0)^* \widehat{\Psi}_N(0)$. It is not discarded in (H15)–(H17). The phase variable is a character of the translation lattice. It is not the coefficient translation a of the exponential note: χ , A , η , and g are fixed throughout (H11)–(H17).

3. Every finite cyclic cover is an exact character decomposition

Let $m \geq 1$ and $\theta_j = 2\pi j/m$. For a function f on the circle of length mL , define

$$Q_j f(r) = \frac{1}{m} \sum_{a=0}^{m-1} e^{ia\theta_j} f(r + aL). \quad (\text{H18})$$

Finite character sums prove $Q_j Q_l = \delta_{jl} Q_j$, $\sum_j Q_j = I$, and $Q_j f(r + L) = e^{-i\theta_j} Q_j f(r)$. The actual unitary map and inverse are

$$f \mapsto (\sqrt{m} Q_j f|_{[0, L]})_{j=0}^{m-1}, \quad f(r + aL) = \frac{1}{\sqrt{m}} \sum_{j=0}^{m-1} e^{-ia\theta_j} g_j(r). \quad (\text{H19})$$

The norm follows either by the finite Fourier identity or orthogonality of the Q_j on the long circle. Applied to the same original source,

$$Q_j \mathcal{Z}_{mL, 0} \psi = \frac{1}{m} \mathcal{Z}_{L, \theta_j} \psi, \quad M_N(mL, 0) = \frac{1}{m} \sum_{j=0}^{m-1} M_N(L, \theta_j). \quad (\text{H20})$$

The second equation is exact at every finite degree; the individual masses are not reset. For a general phase θ , the compatible refinement is

$$M_N(L, \theta) = \frac{1}{n} \sum_{a=0}^{n-1} M_N(L/n, (\theta + 2\pi a)/n). \quad (\text{H21})$$

It follows equally by partitioning the sampled frequency lattice into congruence classes. Repeated refinements compose because these are literal partitions of the same lattice.

The finite locally free coefficient cover

Here is the algebraic cover that gives the finite characters, rather than an identification with an arithmetic Frobenius. Let

$$A_{\text{cov}} = \mathbb{C}[z, z^{-1}], \quad B_{\text{cov}} = \mathbb{C}[w, w^{-1}], \quad z \mapsto w^m. \quad (\text{H22})$$

The pushforward coefficient module is free with its full basis $1, w, \dots, w^{m-1}$. Division of an integer exponent by m proves this also for negative powers. For $\zeta = e^{2\pi i/m}$, the deck action is $\gamma_a(w) = \zeta^{-a}w$. The projector $m^{-1} \sum_a \zeta^{ja} \gamma_a$ has image $A_{\text{cov}} w^j$; multiplication of those summands retains the extra factor z when indices cross m .

The derivative mw^{m-1} is invertible. This proves this particular cover is finite étale, hence finite locally free. EGA I, 1.6.3 and 9.2, and EGA II, 6.1 describe the affine direct-image and finite-morphism setting; here the module and projectors are written explicitly. The base-change map is

$$B_{\text{cov}} \otimes_{A_{\text{cov}}} A' \xrightarrow{\sim} A'[w, w^{-1}]/(w^m - z'), \quad b \otimes a' \mapsto a'b(w). \quad (\text{H23})$$

The circle restriction is $w = e^{-2\pi i r/(mL)}$, $z = e^{-2\pi i r/L}$ and yields (H18). Applying G to the ring map in (H22) commutes with (H3), and fixes τ and e . Character projectors act linearly on the coefficient fibres. A coefficient projected to zero is a represented fibre zero. These idempotent endomorphisms are not relabelled as the supported-zero scalar e .

This is a use of the finite-cover and same-image architecture, not a proof of arithmetic weight bounds from finite étaleness. No six-functor theorem for the split semiring category is asserted.

4. Common arithmetic representatives and the exact descent cost

Suppress N for this section. Let $J : P_N \rightarrow E$ be the original surjection, let $B : \mathbb{C}^r \hookrightarrow P_N$ be the coefficient matrix of χP_{N-q} , and retain $M = \int M_\theta$. For each phase and the common source, define

$$G = (JM^{-1}J^*)^{-1}, \quad C = M^{-1}J^*G, \quad G_\theta = (JM_\theta^{-1}J^*)^{-1}, \quad C_\theta = M_\theta^{-1}J^*G_\theta. \quad (\text{H24})$$

These formulas hold at $N = q - 1$ too: the relation column space then has dimension zero and the corresponding correction is zero. The empty-packet coefficient module has empty Gram determinant one and is handled separately; it does not erase the nonzero analytic seed.

The defining equations are

$$JC = JC_\theta = I_E, \quad B^*MC = 0, \quad B^*M_\theta C_\theta = 0. \quad (\text{H25})$$

Consequently there is a unique coefficient map $X_\theta : E \rightarrow \mathbb{C}^r$ with

$$C - C_\theta = BX_\theta, \quad X_\theta = (B^*M_\theta B)^{-1}B^*M_\theta C. \quad (\text{H26})$$

The last formula follows by applying B^*M_θ to the first. Thus every discrepancy is in the original relation submodule, and monic division followed by the existing tensor theta primitive realizes it in the original cochain complex.

Set

$$\overline{G} = \int G_\theta \frac{d\theta}{2\pi}, \quad \mathcal{D} = \int X_\theta^* (B^* M_\theta B) X_\theta \frac{d\theta}{2\pi}. \quad (\text{H27})$$

The fundamental identity is

$$\boxed{G = \overline{G} + \mathcal{D}.} \quad (\text{H28})$$

Proof. Write $C = C_\theta + BX_\theta$ and expand $C^* M_\theta C$. Both cross terms vanish by (H25), leaving $G_\theta + X_\theta^* B^* M_\theta B X_\theta$. Integrating gives $C^* M C = G$. Every integrand is positive semidefinite. Uniform source bounds justify the finite-dimensional integrals and their inverses; for continuous positive definite matrices on the compact phase circle these bounds also exist without a specified value.

Equation (H28) is the exact difference between first taking the common-source quotient and first minimizing separately in each phase. It replaces a proposed interchange of those operations with a calculated relation norm.

The cochain comparison and its full contractible complement

For m phases put

$$\mathcal{P}_{N,m}^{\text{eq}} = \{(P_j) \in P_N^m : JP_j \text{ is independent of } j\}. \quad (\text{H29})$$

There are actual complexes in degrees 0, 1,

$$[B_N \hookrightarrow P_N] \xrightarrow{\Delta_m} [B_N^m \hookrightarrow \mathcal{P}_{N,m}^{\text{eq}}], \quad (\text{H30})$$

where Δ_m repeats every coefficient. The reverse cochain map is $\text{av}_m(P_j) = m^{-1} \sum_j P_j$ in each degree. It satisfies $\text{av}_m \Delta_m = I$. The complement is

$$\mathcal{B}_{N,m}^0 = \{(b_j) \in B_N^m : \sum_j b_j = 0\}, \quad [\mathcal{B}_{N,m}^0 \xrightarrow{I} \mathcal{B}_{N,m}^0]. \quad (\text{H31})$$

The inverse coefficient coordinates are $P_j \mapsto (\overline{P}, (P_j - \overline{P})_j)$ and $(P, (b_j)) \mapsto (P + b_j)_j$. Since all jets agree, each difference lies in B_N . They prove the asserted direct-sum decomposition of complexes, and the identity is the explicit contraction on the complement. Both complexes have the same degree-one E , with original arithmetic inclusion η . In original arithmetic degree k the complement has degrees $k - 1, k$, so an endomorphism preserving this decomposition has its two complement traces with opposite cohomological signs.

The maps are strict for the stated polynomial filtration: repeating a polynomial does not lower its degree, and averaging a degree- N tuple remains degree at most N . More explicitly, $\Delta_m(P_N) = \text{im } \Delta_m \cap P_N^m$, and averaging the equal-jet tuples at degree N is onto P_N . The multiplication map $M_S : P_N \rightarrow P_{N+1}$ commutes with both operations; its quotient is the original A . At a fixed cutoff it is a degree-raising map, not a presumed endomorphism of P_N .

The complement's cohomology is zero, but its source norm is present in (H27)–(H28). Its contraction is not asserted isometric. Split reconstruction retains its represented vectors, the corresponding supported-zero maps, and external absence. Thus the exactness on the common image does not supply a missing norm estimate by itself.

5. Sharp quadratic control of the descent cost

Theorem. Let $M_\theta \succ 0$ be the specified source matrices with mean M , and suppose

$$(1 - \eta)M \preceq M_\theta \preceq (1 + \eta)M, \quad 0 \leq \eta < 1. \quad (\text{H32})$$

Then for the original J and (H24),

$$\boxed{(1 - \eta^2)G \preceq \overline{G} \preceq G, \quad 0 \leq \mathcal{D} \preceq \eta^2 G.} \quad (\text{H33})$$

Proof of the upper inequality. It is (H28).

Proof of the lower inequality. For $x \in [1 - \eta, 1 + \eta]$, the complete scalar identity is

$$\frac{2 - x}{1 - \eta^2} - \frac{1}{x} = \frac{\eta^2 - (x - 1)^2}{x(1 - \eta^2)} \geq 0. \quad (\text{H34})$$

The endomorphism $X_\theta = M^{-1}M_\theta$ is positive self-adjoint in the retained M -metric and has that spectral interval. Applying (H34) to this finite operator and composing with the displayed metric-dual map M^{-1} proves

$$M_\theta^{-1} \preceq \frac{2M^{-1} - M^{-1}M_\theta M^{-1}}{1 - \eta^2}. \quad (\text{H35})$$

Averaging retains $\int M_\theta = M$ and gives

$$\overline{K} := \int JM_\theta^{-1}J^* \frac{d\theta}{2\pi} \preceq (1 - \eta^2)^{-1}K, \quad K = JM^{-1}J^* = G^{-1}. \quad (\text{H36})$$

For each phase the block matrix

$$\begin{pmatrix} K_\theta & I_E \\ I_E & G_\theta \end{pmatrix} \succeq 0 \quad (\text{H37})$$

is positive: its quadratic form is $(x + G_\theta y)^* K_\theta (x + G_\theta y)$. Averaging and taking the Schur complement gives $\overline{G} \succeq \overline{K}^{-1}$. Inverting (H36) proves the lower inequality of (H33). This proof does not change the original metric, discard the mean condition, or choose a new section to fit a desired form.

More generally, $aM \preceq M_\theta \preceq bM$, $0 < a \leq 1 \leq b$, gives

$$\overline{G} \succeq \frac{ab}{a + b - 1}G, \quad G - \overline{G} \preceq \frac{(1 - a)(b - 1)}{a + b - 1}G. \quad (\text{H38})$$

The inverse chord is $x^{-1} \leq (a + b - x)/(ab)$ on $[a, b]$; the same proof applies.

These are variational positive-operator calculations related to the classical shorted-operator construction. The contribution here is the exact supported source, common jet map, family of theta primitives, and phase recovery to which they are applied.

The arithmetic hypothesis is supplied by the existing correlation bound

For $a > 0$, retain the two actual weighted Grams

$$M_{a,\pm} = (e^{\pm ar}\Psi_N)^*(e^{\pm ar}\Psi_N), \quad \kappa_{a,N} = \sup_{v \neq 0} \frac{v^*(M_{a,+} + M_{a,-})v}{v^*Mv}. \quad (\text{H39})$$

Their original coordinate factors are $x_k^{\pm 2a}$. With $\mathcal{C}(s) = \int \Psi_N(r)^* \Psi_N(r + s)dr$, unfolding gives

$$M_N(\theta) = \sum_{b \in \mathbb{Z}} e^{ib\theta} \mathcal{C}(bL). \quad (\text{H40})$$

Weighted Cauchy–Schwarz bounds each quadratic correlation by $e^{-a|s|} \sqrt{(v^*M_{a,+}v)(v^*M_{a,-}v)}$. Summing both signs proves, uniformly in phase,

$$-\delta_L M \preceq M_N(\theta) - M \preceq \delta_L M, \quad \delta_L = \frac{\kappa_{a,N}}{e^{aL} - 1}. \quad (\text{H41})$$

Thus any specified L with $\delta_L < 1$ supplies (H32). The passage from source error to the averaged quotient error is now quadratic: $\mathcal{D} \preceq \delta_L^2 G$. Growth of $\kappa_{a,N}$ with the arithmetic packet is not bounded by (H41).

6. Endpoints and a genuinely finite phase construction

Apply the uniform phase bound to one finite source containing all four endpoint spaces, for example the original degree $2q + 2$ source. Restriction gives the same η on every smaller source. Write $V_N = \det G_N$ and $\bar{V}_N = \det \bar{G}_N$.

Equation (H33) proves

$$q \log(1 - \eta^2) \leq \log \bar{V}_N - \log V_N \leq 0. \quad (\text{H42})$$

For $\mathcal{B} = \log(V_{q-1}V_q/(V_{2q-1}V_{2q}))$ and its averaged-metric analogue $\bar{\mathcal{B}}$,

$$|\bar{\mathcal{B}} - \mathcal{B}| \leq 2q \log \frac{1}{1 - \eta^2}. \quad (\text{H43})$$

The complete equality remains available:

$$V_N = \bar{V}_N \det(I + \bar{G}_N^{-1} \mathcal{D}_N). \quad (\text{H44})$$

The endomorphism $\bar{G}_N^{-1} \mathcal{D}_N$ is positive self-adjoint for $\bar{G}_N$. Both its source map and its determinant contribution are specified. In particular, averaging the quotient metrics without the correction in (H44) is not an asserted exact reconstruction.

At m equally spaced phases define

$$M^{(m)} = \frac{1}{m} \sum_j M(L, \theta_j) = M(mL, 0), \quad G^{(m)} = (J(M^{(m)})^{-1} J^*)^{-1}, \quad \bar{G}^{(m)} = \frac{1}{m} \sum_j G(L, \theta_j). \quad (\text{H45})$$

Every formula (H24)–(H28) holds exactly with this finite mean and its actual canonical section. To compare this finite construction with the original M , put

$$\delta_m = \frac{\kappa_{a,N}}{e^{amL} - 1}, \quad \nu_m = \frac{\delta_L + \delta_m}{1 - \delta_m}. \quad (\text{H46})$$

For $\delta_m < 1$ and $\nu_m < 1$, triangle inequalities relative to M and $M^{(m)}$ give

$$(1 - \nu_m)M^{(m)} \preceq M(L, \theta_j) \preceq (1 + \nu_m)M^{(m)}. \quad (\text{H47})$$

The theorem and the original quotient transport then prove

$$(1 - \nu_m^2)(1 - \delta_m)G \preceq \bar{G}^{(m)} \preceq (1 + \delta_m)G. \quad (\text{H48})$$

Consequently the finite-phase four-volume error is at most

$$2q \log \frac{1 + \delta_m}{(1 - \delta_m)(1 - \nu_m^2)}. \quad (\text{H49})$$

For fixed sufficiently large L , δ_m tends to zero exponentially with the number of phases. The remaining comparison cost tends to the proved quadratic bound, rather than to an unproved zero.

Exponentially convergent quadrature for the quotient itself

The exact finite-cover identity controls the source mean. There is also a direct, quantitative convergence estimate for the mean of the quotient metrics, with the original matrix inversions included.

Let $\kappa_{\pm}$ be the individual largest generalized eigenvalues of $(M_{a,\pm}, M)$. Weighted Cauchy–Schwarz on two distinct vectors proves

$$\|M^{-1}\mathcal{C}(nL)\|_M \leq \sqrt{\kappa_+ \kappa_-} e^{-a|n|L}. \quad (\text{H49a})$$

Here $\|\cdot\|_M$ is the operator norm in the original source metric. Choose $s > 0$ with $s < aL$ and retain

$$d_s = \frac{2\sqrt{\kappa_+\kappa_-}}{e^{aL-s} - 1} < 1. \quad (\text{H49b})$$

The correlation series defines a holomorphic matrix $\widetilde{M}(z)$ on $|\Im z| < aL$, agreeing with M_θ on the real axis, and $\|M^{-1}(\widetilde{M}(z) - M)\|_M \leq d_s$ on $|\Im z| \leq s$. This is a holomorphic extension of the real-axis Gram coefficients. It is not defined by conjugating the complex parameter in a new Gram formula.

Retain the common section C and original relation columns B from (H24), and define its holomorphic Schur complement

$$\widetilde{G}(z) = C^* \widetilde{M}(z) C - C^* \widetilde{M}(z) B (B^* \widetilde{M}(z) B)^{-1} B^* \widetilde{M}(z) C. \quad (\text{H49c})$$

The inverse exists throughout that strip: its difference from $B^* M B$ has relative operator norm at most $d_s < 1$, so the Neumann series converges. On the real axis (H49c) is exactly G_θ . Orthogonality $B^* M C = 0$ gives diagonal bound $1 + d_s$, cross bounds d_s , and inverse bound $(1 - d_s)^{-1}$ in the stated source/quotient metrics. Consequently

$$\|G^{-1} \widetilde{G}(z)\|_G \leq 1 + d_s + \frac{d_s^2}{1 - d_s} = \frac{1}{1 - d_s}. \quad (\text{H49d})$$

Shift the contour in its periodic Fourier coefficient to $\Im z = -s$ for positive frequency and $+s$ for negative frequency. The two vertical sides cancel by periodicity. This proves the coefficient bound $(1 - d_s)^{-1} e^{-s|n|}$. Root-of-unity averaging keeps only multiples of m , so summing those remaining coefficients gives

$$\|G^{-1}(\overline{G}^{(m)} - \overline{G})\|_G \leq \varepsilon_{m,s} := \frac{2}{(1 - d_s)(e^{sm} - 1)}. \quad (\text{H49e})$$

Both real matrices are Hermitian. Combining this norm inequality with the proved quadratic comparison gives the improved finite statement

$$(1 - \eta^2 - \varepsilon_{m,s})G \preceq \overline{G}^{(m)} \preceq (1 + \varepsilon_{m,s})G. \quad (\text{H49f})$$

It is useful whenever its displayed lower factor is positive. Unlike a bare claim of convergence of phase sampling, (H49b)–(H49f) specify the source-weight cost, the analytic strip, the matrix inverse domain and the complete geometric-series tail. The finite four-volume error is then at most

$$2q \log \frac{1 + \varepsilon_{m,s}}{1 - \eta^2 - \varepsilon_{m,s}}. \quad (\text{H49g})$$

The same maximal finite source supplies constants for all four endpoint restrictions. No off-axis value of ξ has been replaced by a phase character; this is a quadrature estimate for the same canonical quotient construction.

Finite frequency truncation with the phase retained

Write $0 \leq \theta < 2\pi$, $p \geq 1$, and

$$H_{p,N} = \int (1 + r^2) (\partial_r^p \Psi_N)^* \partial_r^p \Psi_N dr. \quad (\text{H50})$$

Integration by parts and Cauchy–Schwarz prove $\|\widehat{\Psi_N v}(u)\|^2 \leq \pi |u|^{-2p} v^* H_{p,N} v$. For an integer $J \geq 2$, retain the frequencies $|n| \leq J$. Since $|n + \theta/(2\pi)| \geq |n| - 1$ on the omitted indices,

$$0 \preceq M(L, \theta) - M(L, \theta; J) \preceq \frac{1}{2p-1} \left(\frac{L}{2\pi(J-1)} \right)^{2p-1} H_{p,N}. \quad (\text{H51})$$

The two tails and the factor π/L combine to give exactly this coefficient. A source-relative upper bound $H_{p,N} \preceq h_{p,N}M$ therefore supplies a uniform omitted error ϵ_J . If $\epsilon_J < 1 - \delta_L$, put $\xi_J = \epsilon_J/(1 - \delta_L)$. Then

$$(1 - \xi_J)G(L, \theta) \preceq G(L, \theta; J) \preceq G(L, \theta). \quad (\text{H52})$$

Combining either (H48) or (H49f) with (H52) multiplies that lower factor by $1 - \xi_J$ and leaves its upper factor unchanged. The four-volume error is respectively (H49) or (H49g) with the extra term $2q \log(1/(1 - \xi_J))$. Rounded arithmetic samples require an additional supplied enclosure. No unevaluated sample or high-degree matrix has been certified by writing these bounds.

7. The original action and relation primitive descend together

Multiplication by S maps P_N to P_{N+1} . If $C_{N,\theta}$ is the actual phase section, monic division defines the unique $Q_{N,\theta}$ by

$$SC_{N,\theta} - C_{N,\theta}A = \chi Q_{N,\theta}. \quad (\text{H53})$$

Applying J_{N+1} proves divisibility; this is a finite polynomial identity. For the actual theta map the right-hand side has the inherited original tensor primitive. Denote its image by $\mathcal{B}_{N,\theta}$. The half-density and twisted periodization commute with $D^{(k)}$ as in (H10), so

$$D_{L,\theta}R_{N,\theta} - R_{N,\theta}A = \mathcal{B}_{N,\theta}, \quad D_{L,\theta} = -\partial_r + k/2. \quad (\text{H54})$$

On the quasi-periodic domain both arguments have boundary multiplier $e^{-i\theta}$, whose modulus is one. Integration by parts therefore has no residual endpoint term. It gives

$$A^*G_{N,\theta} + G_{N,\theta}A - kG_{N,\theta} = -(R_{N,\theta}^*\mathcal{B}_{N,\theta} + \mathcal{B}_{N,\theta}^*R_{N,\theta}). \quad (\text{H55})$$

The direct integral and finite character maps carry these identities with all phases and source primitives present. They do not make the theta boundary orthogonal to the arithmetic representative. The circle spectrum alone does not settle (H55).

The period differential from the uploaded residue note is compatible with this same A : $\Pi' = -\Pi(A + t\mathcal{R})/u$, $\mathcal{R} = 1_E \otimes \ell$, and

$$u^2\Pi'' + ku\Pi' + u\Pi\mathcal{R} = \Pi((A + t\mathcal{R})^2 - k(A + t\mathcal{R})). \quad (\text{H56})$$

The extra $u\Pi\mathcal{R}$ is retained. The new phase parameter does not replace t or introduce a translated cyclic polynomial.

8. Quantitative transport of the new residue-constituent coefficient

The uploaded note proves the following on the marked cyclic fibre. Put

$$\ell([P]) = [S^{q-1}] \text{rem}_\chi P, \quad \beta(x, y) = \ell(xy), \quad \mathcal{R} = 1_E \otimes \ell. \quad (\text{H57})$$

The residue form β is perfect, with determinant $(-1)^{q(q-1)/2}$. The maps $A^i\mathcal{R}A^j = (A^i1) \otimes (\ell A^j)$ span all endomorphisms. Thus for a fixed nonzero proper A -invariant inclusion $\iota : F \hookrightarrow E$, both $\pi 1_E$ and $\ell_F = \ell\iota$ are nonzero, and

$$\pi\mathcal{R}\iota = (\pi 1_E) \otimes \ell_F : F \rightarrow E/F \quad (\text{H58})$$

has rank exactly one. These are inherited results, including repeated-root primary algebras.

For a positive form M on this same E , retain the coefficient

$$\mathfrak{c}_F(M) = \left(\min_{f \in F} \|1_E - \iota f\|_M^2 \right) \ell_F(\iota^* M \iota)^{-1} \ell_F^*. \quad (\text{H59})$$

Apply (H33), without averaging this nonlinear coefficient. Minimizing the first factor and inverting the restriction in the second factor prove

$$(1 - \eta^2) \mathfrak{c}_F(G_N) \leq \mathfrak{c}_F(\overline{G}_N) \leq (1 - \eta^2)^{-1} \mathfrak{c}_F(G_N). \quad (\text{H60})$$

In logarithms the absolute error is at most $-\log(1 - \eta^2)$, with no factor q . For the finite construction, if $aG \preceq M_0 \preceq bG$ are the explicit bounds from (H48), (H52), the factors are a/b and b/a .

For $H(t) = \Pi(t)^* \Pi(t)$ the inherited period curvature is

$$\partial_t \partial_{\bar{t}} \log \det(\iota^* H(t) \iota) = \frac{|t|^2}{|u|^2} \mathfrak{c}_F(H(t)). \quad (\text{H61})$$

Its original first real derivative, for real $u > 0$, is

$$-u \frac{d}{dt} \log \frac{\det(\iota^* H(t) \iota)}{\det(\iota^* G_N \iota)} \Big|_{t=0} - k \dim F = 2\Re \operatorname{Tr} A_F - k \dim F. \quad (\text{H62})$$

Equations (H60)–(H62) connect the new computable source average with the inherited curvature coefficient and the same marked trace observable. They do not convert control of a single coefficient into an upper bound for every endpoint volume. The original period/source condition number is still needed to compare $H(t)$ and G_N .

The translation note's exact map is retained:

$$\Pi_a = f_a \Pi C_a, \quad f_a = \exp((- \Phi(-a) - ta)/u), \quad C_a A_a C_a^{-1} = A + aI. \quad (\text{H63})$$

Its nonzero holomorphic factor leaves the curvature in (H61) unchanged but changes the additive trace in (H62) by $2 \dim F \Re a$. No such translation is performed here; the marked value is $a = 0$. On a finite-field specialization the separate constant character line is the one in the supplied note, not an identification with the additive shift.

9. What premature completion would forget

For a fixed phase, the predecessor's discrete-completion argument gives the map

$$E \longrightarrow \ell^2(\nu_{L,\theta}) / \overline{\chi \mathbb{C}[S]} \simeq \bigoplus_{\chi(k/2 + i(2\pi n + \theta)/L) = 0} \mathbb{C}, \quad (\text{H64})$$

with kernel $(p_{L,\theta})/(\chi)$, where $p_{L,\theta}$ is the squarefree product over the sampled roots. This follows from polynomial density for the exponentially weighted discrete measure and its $|\chi|^2$ multiple; it is a further Hilbert quotient, not (H24).

Only finitely many phases modulo 2π can sample a root of the fixed polynomial. Thus the direct integral of the **completed quotient targets** in (H64) is zero almost everywhere, even when the original E is nonzero. Its map from E has kernel all of E . In split coordinates that map sends represented inputs to the zero of their receiving support; it still fixes external absence. At every finite degree in (H24), $J C_\theta = I_E$ and the averaged Gram is positive definite.

The typed order of operations is therefore part of the result: the exact source isometry (H13), the original finite quotient (H24), and the later completion (H64) are accompanied by their actual maps and kernels. None is silently exchanged with another. The contractible complement in (H31) and the positive norm (H28) likewise coexist.

10. Exact calibrations and the remaining arithmetic estimate

The constant in (H33) is sharp. Take a coefficient source with the literal mass 7,

$$M_{\pm} = 7 \begin{pmatrix} 1 & \pm\eta \\ \pm\eta & 1 \end{pmatrix}, \quad M = \frac{1}{2}(M_+ + M_-) = 7I_2, \quad J = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \quad (\text{H65})$$

These are also moment matrices of the two measures on the points $-1, 1$ with masses $7(1 \mp \eta)/2$ and $7(1 \pm \eta)/2$. Direct minimization gives

$$C = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad C_+ = \begin{pmatrix} 1 \\ -\eta \end{pmatrix}, \quad C_- = \begin{pmatrix} 1 \\ \eta \end{pmatrix}, \quad G = 7, \quad G_{\pm} = 7(1 - \eta^2), \quad \mathcal{D} = 7\eta^2. \quad (\text{H66})$$

Every difference is in the original kernel of J . This is a finite positive-source calibration, not a zeta packet. It proves that the quadratic loss cannot be reduced under only (H32) and the mean condition.

The new exact checker includes repeated-root polynomial quotient fixtures, complex Gram cross terms, finite character projectors, the cochain complement and contraction, source-mean equality, and the precise residue-coefficient comparison. It uses exact rational or Gaussian-rational arithmetic. Its finite checks do not certify the infinite theta integrals, the period determinant argument, or a general Lean theorem.

The arithmetic endpoint issue remains the inherited threshold-four problem for the fixed hypothetical quartet: the original source would require $\mathcal{B}_{h,k} \geq 4q_k \log(D_h k)$. Equation (H43) and its finite version transport that threshold with an explicit $O(q_k)$ allowance when the supplied error factors are fixed. They do not prove a contradicting upper bound for $\bar{\mathcal{B}}_{h,k}$.

The progress here is a stronger, source-preserving descent: all holonomies recover the source exactly; finite covers assemble those holonomies coherently; the failure of independently minimized local representatives to glue is an explicit theta-relation norm; its effect is quadratically bounded; and the same bound reaches the new residue-curvature coefficient. The original complex, infinite arithmetic quotient, support labels, arithmetic action, and full jets remain throughout.

Source basis and attribution

The owner supplies the SplitZero, theta, canonical full-jet, periodization, and residue-constituent programme. This note supplies the displayed all-holonomy comparison, exact common-versus-local minimum correction, its sharp quadratic enclosure, and the explicit joins to that programme. No global novelty claim is made for Fourier-series decomposition, finite-cover character projectors, positive-matrix concavity, or Schur complements.

The new uploaded primary-source correction is accepted at its stated review scope. This session read its TeX, not the underlying scan anew. The selected uploaded EGA readings were I, 1.6.3 and 9.2.1–9.2.2, and II, 6.1.1–6.1.5; neither the entire EGA corpus nor the complete Deligne corpus has been audited. The explanatory Deligne sidebar’s comparison, cone, dual, and split-lift sections were read. The standalone `Deligne_Split_Cohomology(2).tex` is a wrapper; its sibling sidebar was read as the intended substantive input without silently editing its historical include path.

Primary external attribution:

- Deligne, P. (1980). *La conjecture de Weil II*. Publications mathématiques de l’IHÉS 52, 137–252. The supplied correction retains the precise roles of 1.8.12, 3.2.10, 3.2.13, 3.3.4–3.3.6 and 3.7.2–3.7.4.
- Grothendieck, A., and Dieudonné, J. *Éléments de géométrie algébrique*, the selected uploaded editable sections identified above. For the finite locally free definition, see also Stacks Project, Tag 02K9; for affine morphisms and their coefficient algebra, Tag 01S5.
- Zak, J. (1967). Finite translations in solid-state physics. *Physical Review Letters* 19, 1385–1387. DOI 10.1103/PhysRevLett.19.1385. The publisher abstract was inspected for historical attribution; (H11)–(H21) have proofs in this note rather than a claim to have audited the full paper.

- Anderson, W. N., Jr., and Trapp, G. E. (1975). Shorted operators II. *SIAM Journal on Applied Mathematics* 28, 60–71. DOI 10.1137/0128007. The publisher abstract and bibliographic page were inspected; the exact finite inequality used here is proved in (H34)–(H38).

Source files, byte hashes, exact selected ranges, execution receipts, and current-repository inspection are recorded separately. Literature source files and the owner’s unrelated corpus are not republished in this delivery.

Descent cost and generator control

Holonomy descent cost and the original arithmetic generator

This calculation joins the complete holonomy source construction to the original metric commutator and residue calculation. We retain the tensor degree $k \in \mathbb{Z}_{\geq 1}$, the monic polynomial $\chi = \chi_{h,k}$ of degree $q \geq 1$, all its primary multiplicities, the coefficient frame of $E = \mathbb{C}[S]/(\chi)$, $A = M_S$, $e_0 = [1]$, and $\ell[P] = [S^{q-1}] \text{rem}_\chi P$. Put $R = e_0 \ell$. The arithmetic inclusion $\eta[P] = v_h^{\otimes k} P(A_k)1$ and the source seed with Mellin transform g/h , $g = 2\xi$, remain fixed. The scalar error below is written δ , so it does not rename that inclusion. The phase parameter θ , period parameter t , and coefficient translation parameter remain separate original coordinates.

The actual relation norm as a metric endomorphism

Fix $N \geq q - 1$ and let $J : \mathcal{P}_N \rightarrow E$ be the original polynomial remainder map. Denote the literal multiplication matrix of χ by $B_{\text{rel}} : \mathbb{C}^{N+1-q} \hookrightarrow \mathcal{P}_N$. For $N = q - 1$ its domain is zero. Let Ψ_N be the original source columns after the half-density map

$$(\mathcal{U}_k F)(r, z) = e^{(kr + \sum_{i < k} z_i)/2} F(e^{r+z_1}, \dots, e^{r+z_{k-1}}, e^r), \quad \mathcal{K} = L^2(\mathbb{R}^{k-1}, dz). \quad (\text{HG1})$$

The logarithmic Jacobian is $d^k x = e^{kr + \sum z_i} dr dz$. Thus the displayed map preserves the original source norm, and $\mathcal{U}_k D^{(k)} = (-\partial_r + k/2) \mathcal{U}_k$. For $L > 0$, define

$$\begin{aligned} (\mathcal{Z}_{L,\theta} \psi)(r) &= \sum_{a \in \mathbb{Z}} e^{ia\theta} \psi(r + aL), \quad 0 \leq r < L, \quad d\mu(\theta) = \frac{d\theta}{2\pi}, \\ M &= \Psi_N^* \Psi_N, \quad M_\theta = (\mathcal{Z}_{L,\theta} \Psi_N)^* (\mathcal{Z}_{L,\theta} \Psi_N). \end{aligned} \quad (\text{HG2})$$

For a square-integrable source, Fourier series in the translation index give $\int \|\mathcal{Z}_{L,\theta} \psi\|^2 d\mu = \|\psi\|^2$. Polarization gives $\int M_\theta d\mu = M$. The inverse is $\psi(r + aL) = \int e^{-ia\theta} \mathcal{Z}_{L,\theta} \psi(r) d\mu$, first as a Fourier coefficient for almost every r , and hence in the original Hilbert space. The actual finite columns have the weighted regularity established in the source construction, so their sums and derivatives converge in the stated norms.

Keep the two weighted source Grams and their actual comparison value

$$M_{a,\pm} = (e^{\pm ar} \Psi_N)^* (e^{\pm ar} \Psi_N), \quad \kappa = \max_{v \neq 0} \frac{v^* (M_{a,+} + M_{a,-}) v}{v^* M v}, \quad \delta = \frac{\kappa}{e^{aL} - 1}, \quad a > 0. \quad (\text{HG3})$$

Their original coordinate weights are $x_k^{\pm 2a}$. For $\mathcal{C}(s) = \int \Psi_N(r)^* \Psi_N(r + s) dr$, weighted Cauchy–Schwarz gives, for $s \geq 0$, $|v^* \mathcal{C}(s) v| \leq e^{-as} \sqrt{(v^* M_{a,-} v)(v^* M_{a,+} v)}$; the bound at $-s$ follows by adjunction. Unfolding the periodization gives $M_\theta = \sum_{b \in \mathbb{Z}} e^{ib\theta} \mathcal{C}(bL)$. Sum the two tails and use $2\sqrt{uv} \leq u + v$ to obtain $-\delta M \preceq M_\theta - M \preceq \delta M$. Choose L as explicitly specified below so that $\delta < 1$. This proves positivity of every phase Gram in the admitted original source, including the cases where individual sampled weights vanish.

Retain its canonical sections and Grams:

$$\begin{aligned} G &= (JM^{-1}J^*)^{-1}, \quad C = M^{-1}J^*G, \\ G_\theta &= (JM_\theta^{-1}J^*)^{-1}, \quad C_\theta = M_\theta^{-1}J^*G_\theta, \quad \overline{G} = \int G_\theta d\mu. \end{aligned} \quad (\text{HG4})$$

Completing the square on $J^{-1}(v)$ proves these are exactly the minimum-norm representatives. In particular $JC = JC_\theta = I$ and $B_{\text{rel}}^* MC = B_{\text{rel}}^* M_\theta C_\theta = 0$. There is a unique X_θ with

$$\begin{aligned} C - C_\theta &= B_{\text{rel}} X_\theta, \quad X_\theta = (B_{\text{rel}}^* M_\theta B_{\text{rel}})^{-1} B_{\text{rel}}^* M_\theta C, \\ \mathcal{D} &= \int (C - C_\theta)^* M_\theta (C - C_\theta) d\mu, \quad G = \overline{G} + \mathcal{D}. \end{aligned} \quad (\text{HG5})$$

The equality follows by expanding $C = C_\theta + B_{\text{rel}}X_\theta$: the two cross terms vanish by the displayed orthogonality, and $\int C^*M_\theta C d\mu = C^*MC = G$. Empty relation columns give $C = C_\theta = J^{-1}$ and $\mathcal{D} = 0$ directly. Every $B_{\text{rel}}X_\theta v$ is multiplication by the original χ ; it has the original monic-division theta primitive in the retained tensor complex. In particular this positive norm is evaluated on actual relations before passing to their cohomology class.

Here is a complete proof of the quadratic bound used below. For $1 - \delta \leq x \leq 1 + \delta$,

$$\frac{2-x}{1-\delta^2} - \frac{1}{x} = \frac{\delta^2 - (x-1)^2}{x(1-\delta^2)} \geq 0. \quad (\text{HG6})$$

Apply the finite spectral theorem to the positive M -self-adjoint endomorphism $M^{-1}M_\theta$ and then the original dual map M^{-1} . This gives $M_\theta^{-1} \preceq (2M^{-1} - M^{-1}M_\theta M^{-1})/(1 - \delta^2)$. Consequently $\bar{K} = \int JM_\theta^{-1}J^*d\mu \preceq G^{-1}/(1 - \delta^2)$. Each block matrix with rows (G_θ^{-1}, I) and (I, G_θ) is positive: its quadratic form at (x, y) is $(x + G_\theta y)^*G_\theta^{-1}(x + G_\theta y)$. Averaging and taking the Schur complement proves $\bar{G} \succeq \bar{K}^{-1} \succeq (1 - \delta^2)G$. Together with (HG5) this proves

$$0 \preceq \mathcal{D} \preceq \delta^2 G, \quad D = G^{-1}\mathcal{D} = D^\dagger G, \quad 0 \preceq_G D \preceq_G \delta^2 I, \quad G^{-1}\bar{G} = I - D. \quad (\text{HG7})$$

The original inner product convention is $\langle v, w \rangle_G = v^*Gw$. No metric or vector mass is replaced in this construction.

The full commutator receives the quadratic relation cost

Put $T_t = A + tR - kI/2$ for the original complex parameter t and real integer k . For any actual positive Gram H on E , let

$$X_H = T_t^{\dagger H} + T_t, \quad \epsilon_H(t) = \|X_H\|_H, \quad B = G^{-1}(H - G). \quad (\text{HG8})$$

The endomorphism B is G -self-adjoint. Suppose its actual spectral lower bound is $I + B \succeq_G \alpha I$ with $\alpha > 0$; the concrete holonomy and finite values of α are proved below. Let $S = (I + B)^{1/2}$ be the positive square root in the G metric. It is the exact isometry $(E, H) \rightarrow (E, G)$ because $S^*GS = G(I + B) = H$. Direct multiplication gives

$$\begin{aligned} SX_H S^{-1} - X_G &= Z + Z^{\dagger G}, \quad Z = [S, T_t]S^{-1}, \\ [S, T_t] &= \int_0^\infty e^{-rS} [B, T_t] e^{-rS} dr. \end{aligned} \quad (\text{HG9})$$

For the first equality expand the left side as $S^{-1}T_t^{\dagger G}S + ST_t S^{-1} - T_t^{\dagger G} - T_t$. Its two ordered terms are exactly $Z^{\dagger G}$ and Z . For the second equality, both sides solve the Sylvester equation $SY + YS = [B, T_t]$. The integral converges since $\|e^{-rS}\|_G \leq e^{-r\sqrt{\alpha}}$. Differentiate the integrand to verify the equation, with its boundary value $[B, T_t]$ at zero and zero at infinity. In the spectral decomposition of S , a homogeneous solution has each block multiplied by a strictly positive sum of two eigenvalues, so is zero. This proves the claimed integral equality, including repeated eigenvalues. Its norm is at most $\|[B, T_t]\|_G/(2\sqrt{\alpha})$. The isometry, reverse triangle inequality, and $\|S^{-1}\|_G \leq \alpha^{-1/2}$ therefore prove

$$|\epsilon_H(t) - \epsilon_G(t)| \leq \|Z + Z^{\dagger G}\|_G \leq \alpha^{-1} \|[B, A] + t[B, R]\|_G. \quad (\text{HG10})$$

This is the full actual-metric transfer supplied by the coordinating Zeta task, here proved for the present maps and parameters.

For the exact phase mean set $H = \bar{G}$. Equations (HG5)–(HG7) give $B = -D$ and $\alpha = 1 - \delta^2$. Hence

$$\begin{aligned} |\epsilon_{\bar{G}}(t) - \epsilon_G(t)| &\leq \|Z + Z^{\dagger G}\|_G \leq \frac{\|[D, A] + t[D, R]\|_G}{1 - \delta^2}, \\ \frac{\|[D, A] + t[D, R]\|_G}{1 - \delta^2} &\leq \frac{\delta^2}{1 - \delta^2} \|A + tR - kI/2\|_G. \end{aligned} \quad (\text{HG11})$$

Here is the full spectral argument for the last inequality. For any G -self-adjoint B with distinct eigenvalues $b_1 < \dots < b_r$, retain its original spectral projections and put $P_j = \mathbf{1}_{(b_j, \infty)}(B)$. The exact decomposition is $B = b_1 I + \sum_{j=1}^{r-1} (b_{j+1} - b_j) P_j$. In the G -orthogonal decomposition into the image and kernel of P_j , the commutator $[P_j, T]$ has block rows $(0, T_{12})$ and $(-T_{21}, 0)$. Its norm is $\max(\|T_{12}\|, \|T_{21}\|) \leq \|T\|_G$, as follows by applying the two off-diagonal blocks to the orthogonal components of an arbitrary vector. Consequently

$$\|[B, T]\|_G \leq (b_r - b_1) \|T\|_G. \quad (\text{HG11a})$$

When $r = 1$, the sum is empty and the commutator is zero. This proof retains the complete eigenvalue data and every spectral subspace. Applying it to (HG7) gives the last inequality in (HG11); the exact commutator in its preceding line remains available. Thus the original source-relative phase error δ reaches the generator through a proved quadratic allowance. If $[\mathbf{D}, T_t] = 0$, the integral in (HG9) is zero and the two control radii agree exactly. At $N = q - 1$ this equality holds for every t because $\mathcal{D} = 0$.

Exact residue contribution from two original Gram determinants

For any G -self-adjoint B in (HG8), retain $x = e_0$ and $y = G^{-1}\ell^*$. Then $\ell(v) = \langle y, v \rangle_G$ and $R = xy^{\dagger G}$. Define the maps $U, V : \mathbb{C}^2 \rightarrow E$ and the six actual scalar moments

$$\begin{aligned} U &= [Bx, x], & V &= [y, -By], & [B, R] &= UV^{\dagger G}, \\ a_0 &= \|Bx\|_G^2, & c_0 &= \|x\|_G^2, & d_0 &= \|y\|_G^2, & f_0 &= \|By\|_G^2, \\ p_0 &= \langle x, Bx \rangle_G \in \mathbb{R}, & r_0 &= \langle y, By \rangle_G \in \mathbb{R}. \end{aligned} \quad (\text{HG12})$$

Self-adjointness proves the two reality statements. The signed factorization follows on a vector v from $Bx\langle y, v \rangle_G - x\langle By, v \rangle_G = Bx\ell(v) - x\ell(Bv)$. Its two coefficient Grams are

$$\begin{aligned} P &= U^{\dagger G} U = \begin{pmatrix} a_0 & p_0 \\ p_0 & c_0 \end{pmatrix}, & Q &= V^{\dagger G} V = \begin{pmatrix} d_0 & -r_0 \\ -r_0 & f_0 \end{pmatrix}, \\ T_{\text{res}} &= \text{tr}(PQ) = a_0 d_0 + c_0 f_0 - 2p_0 r_0, & \Delta_{\text{res}} &= (a_0 c_0 - p_0^2)(d_0 f_0 - r_0^2). \end{aligned} \quad (\text{HG13})$$

Both negative cross entries and their complete contribution remain. Let $K = UV^{\dagger G}$. The nonzero eigenvalues of $K^{\dagger G} K = V P V^{\dagger G}$, with multiplicities, are those of PQ . To see this without a rank assumption, compute the determinant of the block matrix with rows $(I, -zA_0)$ and (B_0, I) by either diagonal block elimination. This proves $\det(I + zA_0 B_0) = \det(I + zB_0 A_0)$ for rectangular maps. Apply it to $A_0 = V$, $B_0 = P V^{\dagger G}$. Applying the same identity in dimension two proves that PQ has the same characteristic polynomial as the positive semidefinite matrix $P^{1/2} Q P^{1/2}$, including when P is singular. Its two eigenvalues are therefore real and nonnegative. Their sum and product are (HG13). It follows that

$$\|[B, R]\|_G^2 = \frac{T_{\text{res}} + \sqrt{T_{\text{res}}^2 - 4\Delta_{\text{res}}}}{2}. \quad (\text{HG14})$$

The square root is the nonnegative real one. The discriminant is nonnegative by the positive spectrum just proved, and both factors of Δ_{res} are nonnegative Gram determinants.

There is no division by a vector mass or determinant. If $T_{\text{res}} = 0$, positivity gives $K = 0$ and $\Delta_{\text{res}} = 0$. If $\Delta_{\text{res}} = 0 < T_{\text{res}}$, the rank is one and its norm is $\sqrt{T_{\text{res}}}$. If $\Delta_{\text{res}} > 0$, the rank is two; a zero discriminant retains both equal singular-value squares $T_{\text{res}}/2$. For $q = 1$, write $B = bI$ with b real. Then $a_0 = b^2 c_0$, $p_0 = b c_0$, $f_0 = b^2 d_0$, $r_0 = b d_0$, which give $T_{\text{res}} = \Delta_{\text{res}} = 0$ with the original masses unchanged. Repeated roots of χ do not enter any inversion in this argument.

For the holonomy application use $B = -\mathbf{D}$ in (HG12)–(HG14). One may bound the full allowance by

$$\frac{\|[\mathbf{D}, A] + t[\mathbf{D}, R]\|_G}{1 - \delta^2} \leq \frac{\|[\mathbf{D}, A]\|_G + |t| \sqrt{(T_{\text{res}} + \sqrt{T_{\text{res}}^2 - 4\Delta_{\text{res}}})/2}}{1 - \delta^2}. \quad (\text{HG15})$$

The left side retains cancellation between the arithmetic and residue commutators. The right side is its triangle-inequality bound, while (HG14) itself is an exact norm. The complete rank-two calculation and degenerate cases incorporate the coordinating Zeta task's residue proof.

Finite phase and frequency values with explicit choices

The preceding quantities also have a finite approximation on the same polynomial source. Let $\kappa_{\pm}$ be the largest generalized eigenvalues of $(M_{a,\pm}, M)$ and put $b_a = \sqrt{\kappa_+ \kappa_-}$. Weighted Cauchy–Schwarz on two distinct vectors proves $\|M^{-1}\mathcal{C}(nL)\|_M \leq b_a e^{-a|n|L}$. For $0 < s < aL$ define

$$d_s = \frac{2b_a}{e^{aL-s} - 1}, \quad \varepsilon = \frac{2}{(1 - d_s)(e^{sm} - 1)}, \quad H_m = \frac{1}{m} \sum_{j=0}^{m-1} G_{2\pi j/m}. \quad (\text{HG16})$$

When $d_s < 1$, the correlation series is holomorphic for $|\text{Im } z| < aL$ and its source-relative difference from M has norm at most d_s on the closed smaller strip. Its holomorphic quotient is the literal Schur complement

$$\tilde{G}(z) = C^* \tilde{M}(z) C - C^* \tilde{M}(z) B_{\text{rel}} (B_{\text{rel}}^* \tilde{M}(z) B_{\text{rel}})^{-1} B_{\text{rel}}^* \tilde{M}(z) C. \quad (\text{HG17})$$

Here the star acts on the fixed coefficient maps; it does not conjugate z . Orthogonality for M gives diagonal bound $1 + d_s$, cross bounds d_s , and inverse bound $(1 - d_s)^{-1}$ in the original quotient and relation metrics, by the Neumann series. Therefore $\|G^{-1}\tilde{G}(z)\|_G \leq 1 + d_s + d_s^2/(1 - d_s) = (1 - d_s)^{-1}$. At real $z = \theta$ this is exactly G_{θ} . Shift the contour defining its n th Fourier coefficient to $\text{Im } z = -s \text{ sign}(n)$. Periodicity cancels the vertical sides and bounds the coefficient by $(1 - d_s)^{-1} e^{-s|n|}$. Finite phase averaging keeps the multiples of m ; summing the nonzero such coefficients proves $\|G^{-1}(H_m - \overline{G})\|_G \leq \varepsilon$.

For integer $p \geq 1$ keep the actual derivative Gram $H_{p,N} = \int (1 + r^2)(\partial_r^p \Psi_N)^* \partial_r^p \Psi_N dr$ and its comparison value $h_p = \max_{v \neq 0} v^* H_{p,N} v / (v^* M v)$. The quasi-periodic Fourier basis is $L^{-1/2} e^{-i(2\pi n + \theta)r/L}$, whose coefficient is $L^{-1/2} \hat{\Psi}_N((2\pi n + \theta)/L)$ for the convention $\hat{\psi}(u) = \int \psi(r) e^{iur} dr$. Integration by parts and weighted Cauchy–Schwarz give $\|\widehat{\Psi_N v}(u)\|^2 \leq \pi |u|^{-2p} v^* H_{p,N} v$. For $J \geq 2$, sum the omitted $|n| > J$ with $|n + \theta/(2\pi)| \geq |n| - 1$. Comparing the sum from J with the integral from $J - 1$ proves the uniform relative source tail bound

$$\epsilon_J = \frac{h_p}{2p-1} \left(\frac{L}{2\pi(J-1)} \right)^{2p-1}, \quad 0 \preceq M_{\theta} - M_{\theta,J} \preceq \epsilon_J M, \quad \xi_J = \frac{\epsilon_J}{1 - \delta}. \quad (\text{HG18})$$

The full $\theta = 0, n = 0$ coefficient remains among the retained terms. When $\xi_J < 1$, minimizing on the same remainder fibres gives $(1 - \xi_J)G_{\theta} \preceq G_{\theta,J} \preceq G_{\theta}$. Consequently the actual finite mean and its constants satisfy

$$H_{m,J} = \frac{1}{m} \sum_j G_{2\pi j/m,J}, \quad \alpha = (1 - \xi_J)(1 - \delta^2 - \varepsilon), \quad \beta = 1 + \varepsilon, \quad (\text{HG19})$$

$$\alpha G \preceq H_{m,J} \preceq \beta G.$$

These constants can be made positive by explicit choices from the same original source. Choose budgets $0 < \delta_0 < 1$, $0 < \varepsilon_0 < 1 - \delta_0^2$, and $0 < \xi_0 < 1$, and set

$$L = a^{-1} \log(1 + \kappa/\delta_0 + 4b_a), \quad s = aL - \log(1 + 4b_a),$$

$$m = \max\{1, \lceil s^{-1} \log(1 + 4/\varepsilon_0) \rceil\}, \quad (\text{HG20})$$

$$J = 2 + \left\lceil \frac{L}{2\pi} \left(\frac{h_p}{(2p-1)\xi_0(1-\delta)} \right)^{1/(2p-1)} \right\rceil.$$

Both weighted Grams are positive on this finite source, so $b_a > 0$. Thus $0 < s < aL$, $d_s = 1/2$, $\delta < \delta_0$, $\varepsilon \leq \varepsilon_0$, and $\xi_J \leq \xi_0$. Substitution proves $\alpha > 0$. These choices use actual weighted source integrals;

they are explicit mathematical values without an assertion that numerical interval enclosures for those values have already been computed.

The relation cost and the finite tail remain separate exact summands:

$$B_{m,J} = G^{-1}(H_{m,J} - G) = -D + E_{m,J}, \quad E_{m,J} = G^{-1}(H_{m,J} - \bar{G}). \quad (\text{HG21})$$

Indeed $H_{m,J} \preceq H_m \preceq \bar{G} + \varepsilon G$, whereas $H_{m,J} \succeq (1 - \xi_J)(\bar{G} - \varepsilon G)$. Since $\bar{G} \preceq G$, these imply $-\rho G \preceq H_{m,J} - \bar{G} \preceq \varepsilon G$, where $\rho = \xi_J + (1 - \xi_J)\varepsilon \geq \varepsilon$ because $\varepsilon < 1$. Hence $\|E_{m,J}\|_G \leq \rho$. Use (HG10) with the proved α in (HG19) to obtain

$$\begin{aligned} |\epsilon_{H_{m,J}}(t) - \epsilon_G(t)| &\leq \alpha^{-1} \| -[D, A] - t[D, R] + [E_{m,J}, A] + t[E_{m,J}, R] \|_G, \\ &\| -[D, T_t] + [E_{m,J}, T_t] \|_G \leq \| [D, T_t] \|_G + (\rho + \varepsilon) \| T_t \|_G. \end{aligned} \quad (\text{HG22})$$

The second line uses (HG11a) with the proved spectral interval $[-\rho, \varepsilon]$ of $E_{m,J}$. The primary full commutator retains cancellation between both sources of discrepancy. Independently (HG19) gives the spectral interval $[\alpha - 1, \beta - 1]$ for $B_{m,J}$. Equation (HG11a) therefore yields the further valid allowance $(\beta - \alpha)\|T_t\|_G/\alpha$. The lesser of independently proved allowances is valid. Formula (HG14) applies without change to the actual $B_{m,J}$, including its finite tail, and computes its entire rank-at-most-two residue commutator.

The same finite observation retains the full action boundary

For each retained phase let $C_{\theta,J}$ be the canonical section for $M_{\theta,J}$. Monic division gives $SC_{\theta,J} - C_{\theta,J}A = \chi Q_{\theta,J}$ in $\mathcal{P}_{N+1}$, with all original coefficients. Apply the source, half-density, twisted periodization and projection to $|n| \leq J$ to these two columns. Denote the resulting representative and boundary by $\mathcal{R}_j, \mathcal{B}_j$. Then

$$\begin{aligned} D_{L,\theta_j}\mathcal{R}_j - \mathcal{R}_jA &= \mathcal{B}_j, \quad \mathcal{R}_j^*\mathcal{R}_j = G_{\theta_j,J}, \\ A^*H_{m,J} + H_{m,J}A - kH_{m,J} &= -\frac{1}{m} \sum_j (\mathcal{R}_j^*\mathcal{B}_j + \mathcal{B}_j^*\mathcal{R}_j). \end{aligned} \quad (\text{HG23})$$

Here $D_{L,\theta_j} = -\partial_r + k/2$ on the quasi-periodic domain. The projection commutes with this operator. Integration by parts has zero boundary difference because both columns have boundary multiplier $e^{-i\theta_j}$ of modulus one. Thus $D^* + D = k$, which proves the second line by inserting the first and expanding both ordered terms. For complex t the complete control form adds $\bar{t}R^*H_{m,J} + tH_{m,J}R$ to its second line. This gives the actual finite phase/frequency observation entering $\epsilon_{H_{m,J}}(t)$ in (HG22), with its theta relation intact.

For a nonzero proper A -invariant inclusion $I : F \hookrightarrow E$, the residue coefficient is $\mathfrak{c}_F(T) = \min_x \|e_0 - Ix\|_T^2 \ell I(I^*TI)^{-1}I^*\ell^*$. Both factors are strictly positive: if $e_0 \in F$, cyclicity gives $F = E$; if $\ell|_F = 0$, invariance would force $\ell A^j|_F = 0$ for every j , and perfection of the residue pairing would give $F = 0$. Minimizing the first factor and inverting the second on the same F shows that $\alpha G \preceq T \preceq \beta G$ gives

$$\frac{\alpha}{\beta} \mathfrak{c}_F(G) \leq \mathfrak{c}_F(T) \leq \frac{\beta}{\alpha} \mathfrak{c}_F(G). \quad (\text{HG24})$$

For $T = \bar{G}$ use $\alpha = 1 - \delta^2$, $\beta = 1$; for $T = H_{m,J}$ use (HG19). The constant quotient, original unit, and original functional in this coefficient are unchanged. The exact period metric $\Pi(t)^*\Pi(t)$ retains its separately proved comparison to these source Grams and its curvature $|t|^2 \mathfrak{c}_F(\Pi^*\Pi)/|u|^2$.

The transfer also reaches the original Laplacian, including its numerical range. For $H = \bar{G}$ or $H = H_{m,J}$, use its actual S, Z from (HG9) and set $e = \epsilon_H(t) + \|Z + Z^\dagger\|_G$. Equation (HG10) proves $e \geq \epsilon_G(t)$. Larger proved allowances from (HG11) or (HG22) may replace the second summand. Keep $\mathcal{L}_t = (A + tR)^2 - k(A + tR) = T_t^2 - k^2I/4$. Then the identity $X_G = T_t^{\dagger G} + T_t$ gives $G\mathcal{L}_t = GX_GT_t - T_t^*GT_t - k^2G/4$. For an arbitrary nonzero original vector v , put $z = v^*G\mathcal{L}_tv/(v^*Gv)$, $r = \|T_tv\|_G/\|v\|_G$, and $a_1 + ib_1 = \langle v, X_GT_tv \rangle_G/(v^*Gv)$. Cauchy-Schwarz gives $a_1^2 + b_1^2 \leq e^2r^2$, while

the exact expansion gives $\operatorname{Re} z = a_1 - r^2 - k^2/4$ and $\operatorname{Im} z = b_1$. Completing the square in $er - r^2$ and then using the exact identity

$$\begin{aligned} e^2 \left(\frac{e^2 - k^2}{4} - \operatorname{Re} z \right) - b_1^2 &= (a_1 - e^2/2)^2 + e^2 r^2 - a_1^2 - b_1^2 \geq 0, \\ \operatorname{Re} z &\leq \frac{e^2 - k^2}{4}, \quad (\operatorname{Im} z)^2 \leq e^2 \left(\frac{e^2 - k^2}{4} - \operatorname{Re} z \right) \end{aligned} \tag{HG25}$$

prove the stated enclosure for every vector in the original metric. It includes every eigenvalue by inserting its original eigenvector, with generalized eigenspaces left in the source. No division by e occurs, so $e = 0$ is included. This carries the actual quadratic holonomy allowance and finite relation boundary into the same Laplacian polynomial; it does not infer a bound on $\epsilon_H(t)$ from its finite presentation alone.

Finally every invertible original coordinate change $v = Wv'$ sends G, H to W^*GW, W^*HW , and sends B, D, T_t to their $W^{-1}(\cdot)W$ matrices. It sends x, y to $W^{-1}x, W^{-1}y$ and ℓ to ℓW . The positive square root in (HG9) transforms by the same similarity by uniqueness. Therefore the six moments, Gram determinants, all norms, and each ordered identity above are unchanged as values of the original objects. On a retained split support label λ , each coefficient map f acts by $(\lambda, v) \mapsto (\lambda, fv)$ and fixes external τ . A vanishing relation or commutator is the receiving supported zero. This applies also to the isometries and inverse maps above, so the support observation and arithmetic observation commute with every displayed comparison.

Exact fixed-phase observation and its degree cost

Fix the original packet h , integer tensor degree $k \geq 1$, and $\chi = \chi_{h,k}$; retain $g = 2\xi$. This section varies only the polynomial cutoff N . The original source density and its measure are

$$w_h(u) = \frac{|(g/h)(1/2 + iu)|^2}{2\pi}, \quad m(u) = w_h^{*k}(u), \quad d\nu(u) = m(u) du, \quad S(u) = k/2 + iu. \quad (\text{HD1})$$

The full gamma/eta estimates in the periodized-source proof give $\int e^{b|u|} w_h(u) du < \infty$ for $0 < b < \pi/2$. For this same b the inequality $e^{b|u_1+\dots+u_k|} \leq \prod_i e^{b|u_i|}$ and Tonelli's theorem give $\int e^{b|u|} m(u) du \leq (\int e^{b|u|} w_h(u) du)^k < \infty$. The measure has no atoms, and its density is positive almost everywhere: g/h has isolated real-line zeros after the removable selected roots are filled by their full derivatives. For $k \geq 2$ convolution is positive at every real argument, as proved by integrating the positive almost-everywhere factors. In particular every finite original polynomial Gram is positive.

Let M_N be that Gram in the exact coefficient frame $1, S, \dots, S^N$. For a fixed real u_0 put $S_0 = k/2 + iu_0$ and $a_N = (1, S_0, \dots, S_0^N)$. Retain the actual values

$$\Lambda_N(u_0) = a_N M_N^{-1} a_N^*, \quad c_N = M_N^{-1} a_N^* / \Lambda_N(u_0), \quad P_N^{u_0}(S) = \sum_{j=0}^N (c_N)_j S^j. \quad (\text{HD2})$$

The superscript labels this particular polynomial, rather than changing the original source space $\mathcal{P}_N$. The Cauchy–Schwarz inequality in the M_N metric gives

$$\Lambda_N(u_0) = \sup_{P \in \mathcal{P}_N \setminus \{0\}} \frac{|P(S_0)|^2}{\int |P(S(u))|^2 d\nu(u)}, \quad (\text{HD3})$$

$$P_N^{u_0}(S_0) = 1, \quad \int |P_N^{u_0}(S(u))|^2 d\nu(u) = \Lambda_N(u_0)^{-1}.$$

Equality in the supremum follows by inserting $M_N^{-1} a_N^*$. This calculation retains every original mass and phase of M_N .

We prove $\Lambda_N(u_0) \uparrow \infty$, including at a root of χ . Monotonicity follows from the nested polynomial spaces. Suppose for contradiction their suprema had finite bound K . Then evaluation at S_0 would be a bounded linear functional on the polynomial subspace of $L^2(\nu)$, and would extend to its closure. The Hilbert space representation theorem supplies f in that closure with $P(S_0) = \int \bar{f}(u) P(S(u)) d\nu(u)$ for every polynomial P . Since $S(u) = k/2 + iu$ is an invertible affine coordinate over $\mathbb{C}$, this includes all polynomials in u with their full coefficients. Define the finite complex measure $d\sigma = \bar{f} d\nu - \delta_{u_0}$. It has all polynomial moments zero. Cauchy–Schwarz gives

$$\int e^{b|u|/2} |f(u)| d\nu(u) \leq \|f\|_{L^2(\nu)} \left(\int e^{b|u|} d\nu(u) \right)^{1/2} < \infty. \quad (\text{HD4})$$

Thus $F(z) = \int e^{izu} d\sigma(u)$ is holomorphic in $|\text{Im } z| < b/2$. On every strictly smaller strip, dominated differentiation follows by absorbing each power of $|u|$ into the remaining exponential margin. All derivatives of F at zero vanish. Its power series and the identity theorem imply $F = 0$ on that connected strip, and in particular on the real axis.

For completeness, this forces the measure itself to vanish. For $t > 0$ use the Gaussian $\gamma_t(x) = (4\pi t)^{-1/2} e^{-x^2/(4t)}$. Its integrable Fourier representation is $\gamma_t(x) = (2\pi)^{-1} \int e^{-t\zeta^2} e^{-i\zeta x} d\zeta$, obtained by the Gaussian integral. Fubini then gives $(\sigma * \gamma_t)(x) = 0$ since $F(\zeta) = 0$. For each compactly supported continuous φ , $\varphi * \gamma_t \rightarrow \varphi$ uniformly: split the convolution into $|y| < \varepsilon$, controlled by uniform continuity, and its Gaussian tail, whose mass tends to zero. Integration against the finite measure σ therefore gives $\int \varphi d\sigma = 0$. The uniqueness of finite regular measures on continuous test functions gives $\sigma = 0$. But $\bar{f}\nu$ has zero mass at the singleton $\{u_0\}$ and δ_{u_0} has mass one there, a contradiction. This proves

$$\Lambda_N(u_0) \longrightarrow \infty, \quad \mathcal{U}_k \mathcal{V} P_N^{u_0} \longrightarrow 0 \text{ in the original source norm,} \quad P_N^{u_0}(S_0) = 1. \quad (\text{HD5})$$

In particular evaluation on the polynomial domain is not closable as an operator from its original Hilbert closure to $\mathbb{C}$: its graph has the limiting pair $(0, 1)$. This conclusion concerns that specific evaluation map; all finite polynomial maps (HD2) remain exact.

Now fix $L > 0$, $0 \leq \theta < 2\pi$, and an integer n_0 with $u_0 = (2\pi n_0 + \theta)/L$ and $m(u_0) > 0$. Its original sampled mass is $\omega_0 = (2\pi/L)m(u_0) > 0$. The exact phase source matrix is

$$(M_{N,\theta})_{ij} = \frac{2\pi}{L} \sum_{n \in \mathbb{Z}} m((2\pi n + \theta)/L) \overline{S((2\pi n + \theta)/L)^i} S((2\pi n + \theta)/L)^j. \quad (\text{HD6})$$

Its n_0 summand gives $M_{N,\theta} \succeq \omega_0 a_N^* a_N$, so

$$\lambda_{\max}(M_N^{-1} M_{N,\theta}) \geq \omega_0 \Lambda_N(u_0) \longrightarrow \infty. \quad (\text{HD7})$$

For every actual weighted source exponent $a > 0$, let $\kappa_{a,N} = \lambda_{\max}(M_N^{-1}(M_{a,+N} + M_{a,-N}))$. The proved correlation comparison $M_{N,\theta} \preceq (1 + \kappa_{a,N}/(e^{aL} - 1))M_N$ does not require that its error be less than one; it follows from summing the same convergent tails. Consequently

$$\boxed{\kappa_{a,N} \geq (e^{aL} - 1)(\omega_0 \Lambda_N(u_0) - 1), \quad \kappa_{a,N} \longrightarrow \infty.} \quad (\text{HD8})$$

The finite original matrix expression (HD2) is an explicit lower certificate for this degree cost. To ensure the correlation error $\kappa_{a,N}/(e^{aL_N} - 1) \leq \delta_0 < 1$ by that estimate, the literal choice must satisfy $L_N \geq a^{-1} \log(1 + \kappa_{a,N}/\delta_0)$. This derives a constraint on these actual correlation constants at fixed h, k ; it does not assert a growth rate with the distinct tensor parameter k , or exclude another arithmetic estimate.

The source isometry over all holonomies is the exact map that relates this divergent single-phase observation to the full source. Equations (HG2) and (HD3) give

$$\int \|\mathcal{Z}_{L,\vartheta} \mathcal{U}_k \mathcal{V} P_N^{u_0}\|^2 d\mu(\vartheta) = \Lambda_N(u_0)^{-1} \longrightarrow 0, \quad \|\mathcal{Z}_{L,\theta} \mathcal{U}_k \mathcal{V} P_N^{u_0}\|^2 \geq \omega_0. \quad (\text{HD9})$$

The finite columns are regular and define the displayed phase values; the full direct integral is an equivalence class modulo null phase sets. Thus these are simultaneous statements about the very same columns, with the original positive sampled mass retained. In particular point evaluation in the phase parameter cannot be extended as a bounded operator from this full source norm through the direct integral.

There is an exact polynomial-quotient criterion as well. Evaluation at S_0 factors through the original $J : \mathbb{C}[S] \rightarrow E$ if and only if $\chi(S_0) = 0$: necessity follows by evaluating χ , and sufficiency follows from divisibility of every difference in $\ker J = (\chi)$. In that case the induced map $E \rightarrow \mathbb{C}$, $[P] \mapsto P(S_0)$, has kernel $(S - S_0)/(\chi)$, where the notation means the image of the ideal $(S - S_0)$ in E . If S_0 has multiplicity r in χ , write $\chi = (S - S_0)^r q_0$ with $q_0(S_0) \neq 0$. The map on its local factor $\mathbb{C}[x]/(x^r)$ is the constant coefficient, with kernel $(x)/(x^r)$; all other primary factors map to zero. The full nilpotents and every cofactor remain in this map's domain and kernel. If $\chi(S_0) \neq 0$, evaluation is still the exact source map in (HD2)–(HD9); its value on χ explains precisely why it does not factor through E .

For $k \geq 2$, every n_0 has positive mass. For $k = 1$, a sampled selected root of the complete packet has positive mass because, if $h = (S - S_0)^r h_0$ and g has its complete order r there, then $(g/h)(S_0) = g^{(r)}(S_0)/(r!h_0(S_0)) \neq 0$. Other zero-mass samples are absent from the receiving discrete Hilbert space and make no claim in (HD7). For the divergence of $\kappa_{a,N}$ alone, one may choose any real u_0 where $m(u_0) > 0$ and then its unique phase modulo 2π for the fixed L ; such a point exists by (HD1). This gives (HD8) without asserting that every one-fold phase has a positive selected root.

Every displayed source, evaluation, quotient and inverse coefficient map lifts on an original split label λ by $(\lambda, v) \mapsto (\lambda, f v)$ and fixes τ . For each retained label the exact inverse images under this lift are

$$\tilde{f}^{-1}(\{(\lambda, 0)\}) = \{(\lambda, v) : v \in \ker f\}, \quad \tilde{f}^{-1}(\{\tau\}) = \{\tau\}. \quad (\text{HD10})$$

Thus the nonzero coefficient vectors in the source kernel remain represented inputs and map to the receiving represented zero. The external absence has its displayed singleton inverse image. No identification of a positive sampled mass, a represented zero, or external absence is used in the degree-cost calculation.

Completed phase quotients and their full kernels

Independent completed-phase quotient review

Date: 2026-09-13. Scope: the polynomial-density, completed-relation, kernel, and direct-integral assertions in H64. No source file was edited; no scans, program tests, Lean runs, or commutator calculations were performed.

Reviewed text and conclusion

The review uses the following complete supplied sources. The original relative source names below are retained for provenance; their delivered locations and full source hashes are recorded in the companion source manifest.

The relevant fixed text is:

- `web_holonomy_descent_delivery/Tau_Holonomy_Descent_Control/NOTE.md`, lines 67–90: the complete selected orders, original cyclic polynomial, and Taylor unit; lines 166–177: H17 and positivity for tensor degree at least two; lines 703–715: H64, the asserted kernel, and the almost-everywhere direct-integral statement.
- `web_holonomy_descent_delivery/Tau_Holonomy_Descent_Control/SOURCE_REVIEW.md`: the source-reading scope was read. This review adds no primary-source scan claim.
- `web_periodized_source_delivery/Tau_Periodized_Source_Control/RESEARCH_NOTE.md`, lines 88–113: the full-order packet and original cyclic source; lines 564–584: P53–P54 and the exponential-tail density argument; lines 586–612: P55–P57, the target metric, and the closure calculation; line 633: the positive-support qualification for tensor degree one.
- `web_cyclic_sum_delivery/Tau_Cyclic_Sum_Control/RESEARCH_NOTE.md`, lines 31–34: the full-order packet and unit $j_h(g/h)$; lines 54–70: the cyclic minimal polynomial and inclusion.
- `web_arithmetic_endpoint_delivery/Tau_Arithmetic_Endpoint_Bounds/RESEARCH_NOTE.md`, lines 95–105: the full-order packet and the entire quotient defining the actual density.

Conclusion. H64 and its kernel are correct for the stated original full-order arithmetic source at every integer tensor degree $k \geq 1$. For a general nonnegative sampled density, its right side must include only positive-mass sampled roots. At $k = 1$, this qualification makes no change to H64 for the actual source: the ideals $(\chi_{h,1}) = (h)$ agree, and every sampled root has positive mass by full-order cancellation with the original leading coefficient a_h retained. The direct-integral target is zero, with source kernel all of E , in both the general positive-support formulation and its arithmetic specialization. The proof below supplies the explicit metric and the density argument that are abbreviated in H64.

1. Coordinates, mass, and exponential moments

Fix the original integer $k \geq 1$, period $L > 0$, phase $0 \leq \theta < 2\pi$, and the original nonempty full-order packet h . Throughout,

$$c = k/2, \quad u_{n,\theta} = \frac{2\pi n + \theta}{L}, \quad S_{n,\theta} = c + iu_{n,\theta}, \quad n \in \mathbb{Z}, \quad (\text{CQ1})$$

$$v_h = g/h, \quad g = 2\xi, \quad w_h(t) = \frac{|v_h(1/2 + it)|^2}{2\pi}, \quad m_{h,k} = w_h^{*k}, \quad \nu_{n,\theta} = \frac{2\pi}{L} m_{h,k}(u_{n,\theta}). \quad (\text{CQ2})$$

No mass is divided out. Put

$$I_\theta = \{n \in \mathbb{Z} : \nu_{n,\theta} > 0\}, \quad \mathcal{H}_\theta = \ell^2(I_\theta, \nu_\theta). \quad (\text{CQ3})$$

The notation $\ell^2(\nu)$ on all integers means this same Hilbert space after quotienting the zero seminorm coordinates. A zero-mass coordinate is not an evaluation functional on that Hilbert space.

The predecessor's actual-source input, explicitly recorded immediately before P54, is that for each $0 < b < \pi/2$,

$$f_b(t) = e^{b|t|} w_h(t) \in L^1(\mathbb{R}) \cap L^\infty(\mathbb{R}). \quad (\text{CQ4})$$

This is an inherited estimate for the specified entire quotient, not an extra assumption about a new density. Its consequence for sampled measures is proved here. In the convolution integral, $|t_1 + \dots + t_k| \leq |t_1| + \dots + |t_k|$, so

$$e^{b|u|} m_{h,k}(u) \leq f_b^{*k}(u) \leq C_{b,h,k}, \quad C_{b,h,k} := \|f_b\|_\infty \|f_b\|_1^{k-1}. \quad (\text{CQ5})$$

The second inequality follows by integrating one bounded factor and the remaining $k-1$ factors in L^1 . It includes $k=1$. For any $0 \leq a < b$, the literal sum satisfies

$$\begin{aligned} \sum_{n \in I_\theta} \nu_{n,\theta} e^{a|u_{n,\theta}|} &\leq \frac{2\pi C_{b,h,k}}{L} \sum_{n \in \mathbb{Z}} e^{-(b-a)|2\pi n + \theta|/L} \\ &= \frac{2\pi C_{b,h,k}}{L} \frac{e^{-(b-a)\theta/L} + e^{-(b-a)(2\pi-\theta)/L}}{1 - e^{-2\pi(b-a)/L}} < \infty. \end{aligned} \quad (\text{CQ6})$$

The two series use $n \geq 0$ and $n \leq -1$, respectively. Formula CQ6 also includes $\theta = 0$ without deleting its zeroth term.

Let $\chi(S) = \sum_{j=0}^q \chi_j S^j$ be the literal monic original polynomial, of degree $q > 0$. For real u ,

$$|\chi(c + iu)| \leq C_\chi (1 + |u|)^q, \quad C_\chi = \sum_{j=0}^q |\chi_j| \max(1, c)^j. \quad (\text{CQ7})$$

Choose $0 < a_1 < a_2 < b$. The continuous function $(1+t)^{2q} \exp(-(a_2 - a_1)t)$ is bounded on $t \geq 0$, since it tends to zero at infinity. Multiplying CQ7 by the exponential and using CQ6 proves

$$\sum_{n \in I_\theta} |\chi(S_{n,\theta})|^2 \nu_{n,\theta} e^{a_1|u_{n,\theta}|} < \infty. \quad (\text{CQ8})$$

Thus the original discrete measure and its $|\chi|^2$ multiple each have a finite positive exponential moment. All their polynomial moments exist.

2. A complete polynomial-density proof on the shifted lattice

The following argument applies to either measure. Retain the shifted lattice CQ1, a subset I of its indices, and positive masses μ_n such that

$$\sum_{n \in I} \mu_n e^{a|u_{n,\theta}|} < \infty \quad \text{for some } a > 0. \quad (\text{CQ9})$$

Suppose f in $\ell^2(I, \mu)$ is orthogonal to all polynomial evaluation vectors. Since $u = (S - c)/i$, the evaluations of polynomials in S and in the real coordinate u span the same subspace. Conjugating the orthogonality equations if necessary gives

$$\sum_{n \in I} f_n \mu_n u_{n,\theta}^j = 0 \quad (j = 0, 1, 2, \dots). \quad (\text{CQ10})$$

Define

$$F(z) = \sum_{n \in I} f_n \mu_n e^{z u_{n,\theta}}, \quad |\Re z| < a/2. \quad (\text{CQ11})$$

For $0 \leq t < a/2$, Cauchy–Schwarz gives

$$\sum_{n \in I} |f_n| \mu_n e^{t|u_{n,\theta}|} \leq \|f\|_{\ell^2(\mu)} \left(\sum_{n \in I} \mu_n e^{2t|u_{n,\theta}|} \right)^{1/2} < \infty. \quad (\text{CQ12})$$

This proves locally uniform absolute convergence on the open strip in CQ11. For every derivative order j and every smaller closed strip, choose t' strictly between its width and $a/2$. The bound $\sup_{x \geq 0} x^j \exp(-(t' - t)x) < \infty$

and CQ12 at t' prove locally uniform absolute convergence of the derivative series. Hence F is holomorphic and CQ10 says that all its derivatives at zero vanish. Its convergent Taylor series is zero on a disk about zero. Holomorphic uniqueness on the connected strip therefore gives $F(z) = 0$ on that entire strip.

There is no need to appeal to an unproved general Fourier-measure uniqueness statement here. CQ12 at $t = 0$ allows termwise integration, and for each $m \in I$,

$$\begin{aligned} 0 &= \frac{1}{L} \int_0^L F(it) e^{-it u_{m,\theta}} dt \\ &= \sum_{n \in I} f_n \mu_n \frac{1}{L} \int_0^L e^{2\pi i(n-m)t/L} dt = f_m \mu_m. \end{aligned} \quad (\text{CQ13})$$

Because $\mu_m > 0$, every coordinate f_m vanishes. The orthogonal complement of the polynomial span is zero, so that span is dense. CQ6 proves density in $\mathcal{H}_\theta$; CQ8 proves density in the Hilbert space for the positive atoms of $|\chi|^2 \nu_\theta$.

3. The closed relation space, exact map, metric, and kernel

Define

$$Z_\theta = \{n \in I_\theta : \chi(S_{n,\theta}) = 0\}, \quad p_\theta(S) = \prod_{n \in Z_\theta} (S - S_{n,\theta}), \quad (\text{CQ14})$$

where an empty product is 1. The sample points are distinct, so Z_θ has at most q members and p_θ is a squarefree divisor of χ .

Let $\mathcal{R}_\theta$ be the closure in $\mathcal{H}_\theta$ of the evaluation vectors of $\chi \mathbb{C}[S]$. Every such vector vanishes at all indices in Z_θ . Each of those coordinates is continuous, since

$$|f_n| \leq \nu_{n,\theta}^{-1/2} \|f\|_{\mathcal{H}_\theta} \quad (n \in Z_\theta). \quad (\text{CQ15})$$

It follows that $\mathcal{R}_\theta$ is contained in the subspace of vectors vanishing on Z_θ . For the reverse inclusion, let f vanish there and set

$$g_n = \frac{f_n}{\chi(S_{n,\theta})} \quad (n \in I_\theta \setminus Z_\theta). \quad (\text{CQ16})$$

This vector is square-integrable for the weighted masses

$$\mu_n = |\chi(S_{n,\theta})|^2 \nu_{n,\theta},$$

because its squared norm is exactly $\sum_{n \in I_\theta \setminus Z_\theta} |f_n|^2 \nu_{n,\theta}$. By Section 2 there are polynomials P_j tending to g in that weighted norm. Multiplication by the literal χ gives

$$\|\chi P_j - f\|_{\mathcal{H}_\theta}^2 = \sum_{n \in I_\theta \setminus Z_\theta} |P_j(S_{n,\theta}) - g_n|^2 |\chi(S_{n,\theta})|^2 \nu_{n,\theta} \longrightarrow 0. \quad (\text{CQ17})$$

At indices in Z_θ , both terms on the left vanish. Consequently

$$\mathcal{R}_\theta = \{f \in \mathcal{H}_\theta : f_n = 0 \text{ for all } n \in Z_\theta\}. \quad (\text{CQ18})$$

Restriction to Z_θ therefore induces the exact isometric isomorphism

$$\mathcal{H}_\theta/\mathcal{R}_\theta \xrightarrow{\sim} \left(\mathbb{C}^{Z_\theta}, \|(v_n)\|^2 = \sum_{n \in Z_\theta} \nu_{n,\theta} |v_n|^2 \right). \quad (\text{CQ19})$$

The quotient norm is exactly the displayed norm: the coordinate projection onto Z_θ is orthogonal, and the complementary projection has range $\mathcal{R}_\theta$.

Since adding χQ to a polynomial changes its evaluation vector by an element of $\mathcal{R}_\theta$, there is a well-defined linear comparison

$$T_\theta : E = \mathbb{C}[S]/(\chi) \longrightarrow \mathcal{H}_\theta/\mathcal{R}_\theta, \quad [P] \longmapsto (P(S_{n,\theta}))_{n \in Z_\theta}. \quad (\text{CQ20})$$

It is surjective. When Z_θ is nonempty, the literal Lagrange polynomials

$$L_n(S) = \prod_{m \in Z_\theta \setminus \{n\}} \frac{S - S_{m,\theta}}{S_{n,\theta} - S_{m,\theta}} \quad (\text{CQ21})$$

map to its coordinate basis; when it is empty, the target is zero. A polynomial vanishes at every distinct point indexed by Z_θ if and only if it is divisible by their squarefree product. Hence the exact kernel is

$$\ker T_\theta = (p_\theta)/(\chi) \subseteq E. \quad (\text{CQ22})$$

This retains the multiplicities in χ . In the primary decomposition of E , each unsampled primary block lies in the kernel; at a sampled root of multiplicity m , the kernel is the maximal ideal $(S - \rho)/(S - \rho)^m$, while its scalar value survives. CQ20 is a linear map to a Hilbert quotient; no multiplication of arbitrary square-summable sequences is required.

4. Why the original arithmetic source needs no extra positivity hypothesis

For $k \geq 2$, the actual entire nonzero function v_h has isolated zeros. Thus $w_h(t) > 0$ off a discrete subset of the real axis. For fixed u , both $w_h(t)$ and $w_h(u - t)$ are positive for almost every t ; their product is integrable by boundedness and integrability from CQ4. Its integral is strictly positive. Thus $m_{h,2}(u) > 0$ for every real u . Inductively, convolving the everywhere-positive $m_{h,k-1}$ with the almost-everywhere-positive nonzero w_h gives $m_{h,k}(u) > 0$. CQ5 makes the integrals finite. Therefore $I_\theta = \mathbb{Z}$ for $k \geq 2$ and CQ14 is exactly the index set displayed in H64.

For $k = 1$, the original cyclic algebra is $\mathbb{C}[S]/(h)$, with generator multiplication by S . Its annihilator ideal is exactly (h) : $h(M_S) = 0$, and any polynomial annihilating M_S also annihilates 1, so is divisible by the unchanged h . Consequently $(\chi_{h,1}) = (h)$. If $\chi_{h,1}$ denotes the monic minimal polynomial and a_h is the original leading coefficient of h , then $\chi_{h,1} = h/a_h$; this identifies the ideal generator and does not replace the original source $v_h = g/h$. Equality $\chi_{h,1} = h$ is asserted only when the source separately fixes $a_h = 1$.

Write the unchanged packet as

$$h(S) = a_h \prod_{\rho \in Z_h} (S - \rho)^{m_\rho}, \quad a_h \neq 0, \quad m_\rho = \text{ord}_\rho g. \quad (\text{CQ23})$$

For a selected root ρ , full-order removal gives the exact nonzero value

$$v_h(\rho) = \frac{g^{(m_\rho)}(\rho)}{h^{(m_\rho)}(\rho)} = \frac{g^{(m_\rho)}(\rho)}{a_h m_\rho! \prod_{\sigma \in Z_h \setminus \{\rho\}} (\rho - \sigma)^{m_\sigma}} \neq 0. \quad (\text{CQ24})$$

This follows by factoring $(S - \rho)^{m_\rho}$ from both g and the unchanged h ; the numerator derivative, original leading coefficient a_h , and every denominator factor are nonzero by their definitions. If a sample satisfies $S_{n,\theta} = \rho$, then $\Re \rho = 1/2$ and

$$\nu_{n,\theta} = \frac{2\pi}{L} w_h(\Im \rho) = \frac{1}{L} |v_h(\rho)|^2 = \frac{|g^{(m_\rho)}(\rho)|^2}{L|a_h|^2(m_\rho!)^2 \prod_{\sigma \in Z_h \setminus \{\rho\}} |\rho - \sigma|^{2m_\sigma}} > 0. \quad (\text{CQ25})$$

Thus every root appearing in H64 has positive mass also at $k = 1$. Other zero-mass lattice coordinates are discarded by the Hilbert-space definition CQ3, but none removes a summand from H64. The full-order hypothesis at NOTE.md line 67 and the preserved unit at lines 85–90 already supply the needed fact. A useful exposition repair is to define I_θ and then state CQ25; it does not change H64’s conclusion for this source.

For comparison, outside this arithmetic domain the omission of positive support can change the theorem. With $\nu_0 = 0$, $\nu_n = e^{-|n|}$ for $n \neq 0$, and $\chi(S) = S - S_{0,\theta}$, Section 3 gives zero completed quotient, whereas a sum over all sampled polynomial roots would give one copy of $\mathbb{C}$. This example establishes why the general formula must use CQ14; it is not a counterexample to the full-order arithmetic result.

5. The direct-integral target and its exact zero map

Retain the fixed polynomial χ and $c = k/2$. Let

$$\mathcal{F} = \{\theta \in [0, 2\pi) : \theta \equiv L\Im \rho \pmod{2\pi} \text{ for some root } \rho \text{ of } \chi \text{ with } \Re \rho = c\}. \quad (\text{CQ26})$$

Each such root specifies exactly one phase in the chosen half-open interval, and then exactly one integer n with $\rho = S_{n,\theta}$. Therefore $\mathcal{F}$ is finite, with at most q elements. For θ outside $\mathcal{F}$, the index set Z_θ is empty, so CQ19 is the zero Hilbert space.

A concrete measurable realization prevents ambiguity about the field of targets. Map CQ19 into the fixed Hilbert space $\ell^2(\mathbb{Z})$ by

$$(v_n)_{n \in Z_\theta} \longmapsto (\mathbf{1}_{n \in Z_\theta} \sqrt{\nu_{n,\theta}} v_n)_{n \in \mathbb{Z}}. \quad (\text{CQ27})$$

This is an isometry onto the range of the diagonal orthogonal projection with diagonal entries $\mathbf{1}_{\nu_{n,\theta} > 0} \mathbf{1}_{\chi(S_{n,\theta})=0}$. Those entries are measurable: the first condition is measurable for the original continuous density, and the second is the zero set of a continuous function of θ . Hence these ranges define a measurable Hilbert field. Their direct integral has norm

$$\int_0^{2\pi} \sum_{n \in Z_\theta} \nu_{n,\theta} |v_n(\theta)|^2 \frac{d\theta}{2\pi}. \quad (\text{CQ28})$$

Every section is zero outside the finite set $\mathcal{F}$; its norm in CQ28 is zero. The direct integral is therefore the zero Hilbert space. For each $[P] \in E$, the evaluation sections of CQ20 are measurable through CQ27 and supported in $\mathcal{F}$. They consequently define the zero class in the direct integral. The induced map and its exact kernel are

$$E \longrightarrow \int_{[0,2\pi)}^\oplus (\mathcal{H}_\theta / \mathcal{R}_\theta) \frac{d\theta}{2\pi} = 0, \quad \ker = E. \quad (\text{CQ29})$$

Individual exceptional fibres can remain nonzero and carry the positive masses in CQ19; a finite set has zero measure for the actual phase measure $d\theta/(2\pi)$. CQ29 proves precisely the completed-target claim at NOTE.md line 715. It does not change the source isometry H13 or the finite quotient H24.

On a retained split support label ℓ , the actual induced map is $(\ell, v) \longmapsto (\ell, 0)$, while external absence maps to external absence. This is the lift of CQ29 by the stated H4; its coefficient kernel is all of E , exactly as CQ29 records.

Review disposition

Accept H64, its displayed squarefree kernel, and the zero direct-integral conclusion for the original full-order packet. The recommended addition is explanatory: define the positive support and retained metric by CQ3, CQ14, CQ19; explicitly record the degree-one identity and positive sampled-root value CQ25. The unrestricted all-atom version is not justified for arbitrary other nonnegative densities. No source edit or mathematical test result is claimed by this review.

Typed source addendum and sharp source realization

Holonomy descent: the inverse domain, positive root masses, and sharp source realization

This addendum retains the original $\chi = \chi_{h,k}$, its quotient $E = \mathbb{C}[S]/(\chi)$, multiplication $A[P] = [SP]$, the original source g/h with $g = 2\xi$, and every measure in (H9)–(H17). It makes the inverse domain in (H49c) precise, proves the positive-support and shifted-density assertions used in (H64), and realizes the sharp quadratic loss with actual compactly supported source columns. The independent proof review identified the analytic-circle calibration. The full source-density proof PSD1–41 is an accompanying dependency.

The correlation strip and the domain of the quotient inverse

Fix the original source map $\Psi : \mathcal{P}_N \rightarrow L^2(\mathbb{R}; \mathcal{K})$, its Gram $M = \Psi^* \Psi \succ 0$, quotient $J : \mathcal{P}_N \twoheadrightarrow E$, and literal relation injection $B : \mathbb{C}^r \hookrightarrow \mathcal{P}_N$. Define

$$G = (JM^{-1}J^*)^{-1}, \quad C = M^{-1}J^*G, \quad M_B = B^*MB. \quad (\text{HDI1})$$

Then $JC = I$, $C^*MC = G$, $B^*MC = 0$. For $r = 0$, the relation metric and its inverse are the unique automorphism of $\{0\}$; every correction factoring through B is the zero map on its stated domain and codomain. For $r > 0$, $M_B \succ 0$. Retain $a > 0$, $M_{a,\pm} = (e^{\pm ar}\Psi)^*(e^{\pm ar}\Psi)$, and

$$\kappa_{\pm} = \sup_{v \neq 0} \frac{v^* M_{a,\pm} v}{v^* M v}, \quad K_a = \sqrt{\kappa_+ \kappa_-}, \quad \kappa = \sup_{v \neq 0} \frac{v^* (M_{a,+} + M_{a,-}) v}{v^* M v}. \quad (\text{HDI2})$$

These retain the factors $x_k^{\pm 2a}$ in the original coordinates. For $\mathcal{C}(v) = \int \Psi(r)^* \Psi(r+v) dr$, weighted Cauchy–Schwarz on two vectors gives

$$|\langle x, \mathcal{C}(v)y \rangle| \leq K_a e^{-a|v|} \|x\|_M \|y\|_M, \quad \|M^{-1}\mathcal{C}(nL)\|_M \leq K_a e^{-a|n|L}. \quad (\text{HDI3})$$

For $v > 0$, write the integrand as e^{-av} times the pairing of $e^{-ar}\Psi(r)x$ with $e^{a(r+v)}\Psi(r+v)y$. Their squared integrals are $x^* M_{a,-} x$ and $y^* M_{a,+} y$. For $v < 0$, use the opposite signs. The dual norm in the retained M -metric then proves the operator inequality. Thus

$$\widetilde{M}(z) = \sum_{n \in \mathbb{Z}} e^{inz} \mathcal{C}(nL), \quad d_s = \frac{2K_a}{e^{aL-s} - 1}, \quad \|M^{-1}(\widetilde{M}(z) - M)\|_M \leq d_s \quad (|\operatorname{Im} z| \leq s < aL). \quad (\text{HDI4})$$

The series and its derivative series converge normally on every closed substrip of $|\operatorname{Im} z| < aL$: powers of $|n|$ are absorbed by the remaining geometric decay. The zero coefficient is M ; summing both nonzero signs proves the displayed bound. This constructs a holomorphic, periodic extension of the real Gram coefficients, without conjugating the complex parameter in a Gram.

Choose $0 < s < aL$ with $d_s < 1$. Put $T = M^{-1}(\widetilde{M} - M)$. The maps and their adjoints are

$$\begin{aligned} C : (E, G) &\rightarrow (\mathcal{P}_N, M), & C^\dagger &= G^{-1}C^*M, & C^\dagger C &= I, \\ B : (\mathbb{C}^r, M_B) &\rightarrow (\mathcal{P}_N, M), & B^\dagger &= M_B^{-1}B^*M, & B^\dagger B &= I. \end{aligned} \quad (\text{HDI5})$$

They are isometries onto orthogonal subspaces by (HDI1). Consequently the four compressions of T by B, C have norm at most d_s , with these domain and target metrics. In particular,

$$B^* \widetilde{M} B = M_B(I + B^\dagger T B), \quad (I + B^\dagger T B)^{-1} = \sum_{j \geq 0} (-B^\dagger T B)^j \quad (\text{HDI6})$$

exists and has norm at most $(1 - d_s)^{-1}$ on the selected closed strip. Continuity of d_s , and its strict inequality, give $s' > s$ with $s' < aL$, $d_{s'} < 1$. Hence the inverse is holomorphic on a neighborhood of the closed strip used by the contour calculation. The phrase *throughout that strip* after (H49c) means this selected strip, not the whole larger correlation strip $|\operatorname{Im} z| < aL$.

The corresponding Schur complement is

$$\tilde{G} = C^* \tilde{M} C - C^* \tilde{M} B (B^* \tilde{M} B)^{-1} B^* \tilde{M} C, \quad \|G^{-1} \tilde{G}\|_G \leq 1 + d_s + \frac{d_s^2}{1 - d_s} = \frac{1}{1 - d_s}. \quad (\text{HDI7})$$

The bound follows from (HDI5)–(HDI6), including the two cross compressions and their actual metrics. On the real axis, subtract $B(B^* M_\theta B)^{-1} B^* M_\theta C$ from C . This remains a section of J and is M_θ -orthogonal to $\ker J$. Adding any relation adds precisely its nonnegative squared norm, so this is the unique minimizing section C_θ , and its Gram is the displayed Schur complement.

For Fourier coefficients $\hat{G}_n = (2\pi)^{-1} \int_0^{2\pi} e^{-in\theta} \tilde{G}(\theta) d\theta$, shift to imaginary part $-s$ if $n > 0$, and $+s$ if $n < 0$. The just-proved neighborhood of holomorphy permits the contour, and periodicity cancels the two vertical sides. The exponential has modulus $e^{-s|n|}$. Therefore

$$\|G^{-1} \hat{G}_n\|_G \leq \frac{e^{-s|n|}}{1 - d_s}, \quad \left\| G^{-1} \left(\frac{1}{m} \sum_{j=0}^{m-1} G_{2\pi j/m} - \bar{G} \right) \right\|_G \leq \frac{2}{(1 - d_s)(e^{sm} - 1)}. \quad (\text{HDI8})$$

Absolute convergence permits averaging term by term. The finite character sum keeps precisely the indices divisible by m ; summing their two nonzero tails proves the second bound.

Explicit finite choices exist. Choose $s_0 > 0$, $0 < d_0 < 1$, $0 < \eta_0 < 1$, $0 < \varepsilon_0 < 1 - \eta_0^2$, and set

$$\begin{aligned} L &= \frac{1}{a} \max \{s_0 + \log(1 + 2K_a/d_0), \log(1 + \kappa/\eta_0)\}, \\ m &= \max \left\{ 1, \left\lceil \frac{1}{s_0} \log \left(1 + \frac{2}{(1 - d_0)\varepsilon_0} \right) \right\rceil \right\}. \end{aligned} \quad (\text{HDI9})$$

The positive weighted source forms give $K_a > 0$, so $aL > s_0$. Monotonicity gives $d_{s_0} \leq d_0$, $\delta_L = \kappa/(e^{aL} - 1) \leq \eta_0$, and error at most ε_0 in (HDI8). With the delivered quadratic theorem,

$$(1 - \eta_0^2 - \varepsilon_0)G \preceq \frac{1}{m} \sum_{j=0}^{m-1} G_{2\pi j/m} \preceq (1 + \varepsilon_0)G. \quad (\text{HDI10})$$

Its lower factor is positive by the actual selection. These are finite source-dependent formulas; they do not bound growth of $M_{a,\pm}$ with the arithmetic packet. The generator companion HG gives the combined phase and frequency budget for its operators.

Selected-root weights and density on the shifted lattice

Fix $L > 0$, $k \geq 1$, and $0 \leq \theta < 2\pi$, and retain

$$u_n = \frac{2\pi n + \theta}{L}, \quad S_n = k/2 + iu_n, \quad \nu_n = \frac{2\pi}{L} m_{h,k}(u_n), \quad I_\theta = \{n : \nu_n > 0\}. \quad (\text{HDI11})$$

The source-density proof PSD1–22 supplies, for each $0 < b < \pi/2$, the actual finite constants and pointwise bound

$$A_b = \|e^{b|\cdot|} w_h\|_\infty, \quad B_b = \|e^{b|\cdot|} w_h\|_1, \quad m_{h,k}(u) \leq A_b B_b^{k-1} e^{-b|u|}. \quad (\text{HDI12})$$

The full PSD source accompanies this addendum; its Gamma factor, leading coefficient, and cancellation calculation are retained. The entire nonzero function g/h has a discrete zero set on the vertical line, so w_h is continuous and positive away from that discrete set. For a fixed u , choose t outside the zero sets of both $w_h(t)$ and $w_h(u - t)$. An interval of strictly positive convolution integrand follows by continuity, so $m_{h,2}(u) > 0$. Induction integrates the preceding everywhere positive convolution against w_h on an interval of positivity. Thus $I_\theta = \mathbb{Z}$ for $k \geq 2$.

For $k = 1$, use I_θ . At a selected root ρ , let d_ρ be its full multiplicity in both g and h . Then

$$\begin{aligned} g(s) &= (s - \rho)^{d_\rho} a_\rho(s), \quad h(s) = (s - \rho)^{d_\rho} b_\rho(s), \quad a_\rho(\rho) b_\rho(\rho) \neq 0, \\ (g/h)(\rho) &= \frac{a_\rho(\rho)}{b_\rho(\rho)} = \frac{g^{(d_\rho)}(\rho)}{h^{(d_\rho)}(\rho)} \neq 0. \end{aligned} \quad (\text{HDI13})$$

The leading coefficient of h stays in b_ρ and its derivative. In the original monic packet convention $\chi_{h,1} = h$. If a nonzero leading coefficient a_h is displayed, the monic minimal polynomial is $a_h^{-1}h$ and its ideal is (h) ; the analytic source remains exactly g/h . Consequently every sampled root $\rho = S_n$ of $\chi_{h,1}$ has the exact positive weight

$$\nu_n = \frac{1}{L} \left| \frac{g^{(d_\rho)}(\rho)}{h^{(d_\rho)}(\rho)} \right|^2 > 0. \quad (\text{HDI14})$$

This is the full-order cancellation fact that justifies every root index printed in (H64), even for $k = 1$.

Here is the density proof with the phase retained. Let $\mu_n > 0$ on $I \subseteq I_\theta$ have $\sum_I e^{c|u_n|} \mu_n < \infty$ for some $c > 0$, and suppose $v \in \ell^2(I, \mu)$ is orthogonal to all polynomial sequences. For $0 < d < c/2$, Cauchy–Schwarz proves

$$\sum_I |v_n| \mu_n e^{d|u_n|} \leq \|v\|_\mu \left(\sum_I e^{2d|u_n|} \mu_n \right)^{1/2} < \infty. \quad (\text{HDI15})$$

Therefore $F(z) = \sum_I \bar{v}_n \mu_n e^{iz u_n}$ is holomorphic for $|\text{Im } z| < c/2$, with all derivative series normally convergent on smaller strips. The powers are absorbed by the remaining exponential margin. The original-coordinate polynomial $((S - k/2)/i)^j$ evaluates to u_n^j , so every derivative at zero vanishes. The identity theorem gives $F = 0$. Extend the coefficients by zero off I . Absolute summability and $u_n - u_m = 2\pi(n - m)/L$ give

$$\frac{1}{L} \int_0^L F(x) e^{-iu_m x} dx = \sum_I \bar{v}_n \mu_n \frac{1}{L} \int_0^L e^{2\pi i(n-m)x/L} dx = \begin{cases} \bar{v}_m \mu_m, & m \in I, \\ 0, & m \notin I. \end{cases} \quad (\text{HDI16})$$

The left side is zero, so $v = 0$. This proves density. Apply it to $\mu = \nu$ on I_θ , and then to $\mu_n = |\chi(S_n)|^2 \nu_n$ where $\chi(S_n) \neq 0$. For both, (HDI12) gives the required exponential moment because a fixed polynomial is absorbed by the difference of two positive exponential exponents.

Put $Z_\theta = \{n \in I_\theta : \chi(S_n) = 0\}$ and $p_\theta = \prod_{n \in Z_\theta} (S - S_n)$. Coordinate evaluation is continuous in $\mathcal{H}_\theta = \ell^2(I_\theta, \nu)$, with bound $|f_n| \leq \nu_n^{-1/2} \|f\|_\nu$. Thus the closure of $\chi \mathbb{C}[S]$ vanishes on Z_θ . Conversely, if f vanishes there, division by $\chi(S_n)$ on the complement puts f/χ in the second weighted space, with exactly the same squared norm. Polynomial approximation there, followed by multiplication by χ , approximates f in $\mathcal{H}_\theta$. Consequently

$$\begin{aligned} \overline{\chi \mathbb{C}[S]} &= \{f \in \mathcal{H}_\theta : f_n = 0 \ (n \in Z_\theta)\}, \\ \mathcal{T}_\theta &:= \mathcal{H}_\theta / \overline{\chi \mathbb{C}[S]} \cong \left(\mathbb{C}^{Z_\theta}, \sum_{n \in Z_\theta} \nu_n |a_n|^2 \right), \\ e_\theta : E &\twoheadrightarrow \mathcal{T}_\theta, \quad [P] \mapsto (P(S_n))_{n \in Z_\theta}, \quad \ker e_\theta = (p_\theta)/(\chi). \end{aligned} \quad (\text{HDI17})$$

The isometry is restriction, the orthogonal representative of the Hilbert quotient. The evaluation map is well-defined since χ vanishes at these nodes. The explicit Lagrange polynomial $\prod_{m \in Z_\theta, m \neq n} (S - S_m)/(S_n - S_m)$ proves surjectivity. Distinctness of the nodes proves the divisibility description of the kernel. The empty product is one and its target is zero. Polynomial cancellation also gives the exact sequence, equivariant for multiplication by S ,

$$0 \longrightarrow \mathbb{C}[S]/(\chi/p_\theta) \xrightarrow{[Q] \mapsto [p_\theta Q]} E \xrightarrow{e_\theta} \mathcal{T}_\theta \longrightarrow 0. \quad (\text{HDI18})$$

No coprimality of $p_\theta, \chi/p_\theta$ is required. At a root λ of multiplicity d_λ , the original primary block is $\mathbb{C}[\epsilon]/(\epsilon^{d_\lambda})$. At a sampled root this map is its constant coefficient, with kernel $\epsilon\mathbb{C}[\epsilon]/(\epsilon^{d_\lambda})$; at an unsampled root the full block is in the kernel. This follows by substituting $S = \lambda + \epsilon$ into evaluation, using the full explicit Chinese-remainder isomorphism in PSD34–35.

For precision about the original source, let $b_n(\theta) = \mathcal{U}_k F_h^{\otimes k}(u_n)$. The source-to-sequence dictionary is

$$\widehat{I_\theta a}(n) = L^{-1/2} a_n b_n(\theta), \quad \|b_n(\theta)\|_{\mathcal{K}}^2 = 2\pi m_{h,k}(u_n), \quad a_n = \sqrt{L} \frac{\langle b_n(\theta), \widehat{f}(n) \rangle}{\|b_n(\theta)\|_{\mathcal{K}}^2} \quad (n \in I_\theta). \quad (\text{HDI19})$$

The Fourier basis is $L^{-1/2} e^{-i(2\pi n + \theta)r/L}$. Partial Plancherel on the retained relative variables gives the middle equality, as proved pointwise in PSD16. Parseval proves the isometry and its inverse on its range. Polynomial density identifies that range with the closed polynomial source. On polynomials the map is exactly $I_\theta(P(S_n)) = \mathcal{Z}_{L,\theta} \mathcal{U}_k \mathcal{V} P$ by the plus-sign Fourier convention. All amplitudes and the factor $L^{-1/2}$ remain in the comparison.

The measurable field can be specified without an unspecified choice of fibre coordinates. Embed $\mathcal{T}_\theta$ in $\ell^2(\mathbb{Z})$ by $a_n \mapsto \sqrt{\nu_n(\theta)} a_n$ on Z_θ , zero elsewhere. Its range projection is

$$Q_\theta = \text{diag} \left(1_{\{\nu_n(\theta) > 0, \chi(S_n(\theta)) = 0\}} \right)_{n \in \mathbb{Z}}. \quad (\text{HDI20})$$

Each entry is Borel, since ν_n and $\chi(S_n)$ are continuous. The sections $Q_\theta e_n$ form a countable measurable generating family; approximating any fixed vector by finitely supported vectors proves strong measurability of $Q_\theta v$. A nonzero entry requires a root ρ with $\text{Re } \rho = k/2$ and $\theta \equiv L \text{Im } \rho \pmod{2\pi}$. There are only finitely many such phases. Thus the decomposable projection $f(\theta) \mapsto Q_\theta f(\theta)$ on $L^2(d\theta/(2\pi); \ell^2(\mathbb{Z}))$ is zero, because its squared norm is an integral supported on a finite set. It follows exactly that

$$\int^\oplus \mathcal{T}_\theta \frac{d\theta}{2\pi} = 0, \quad E \longrightarrow \int^\oplus \mathcal{T}_\theta \frac{d\theta}{2\pi}, \quad [P] \mapsto [e_\theta[P]]_{\text{a.e.}}, \quad \ker = E. \quad (\text{HDI21})$$

The section in this map is Borel by its explicit coordinates $1_{Z_\theta}(n) \sqrt{\nu_n(\theta)} P(S_n(\theta))$, and has zero norm. Its pointwise exceptional maps remain (HDI17)–(HDI18) with their positive weights; passage to the almost-everywhere class is the exact arrow removing the measure-zero section. The accompanying continuous-cochain comparison further records this operation on its source complex.

On every original support label ℓ , each coefficient map above lifts as $(\ell, v) \mapsto (\ell, f v)$, with $\tau \mapsto \tau$. In particular (HDI21) takes a represented input to the zero in its receiving fibre, with represented support still present, while external absence remains fixed. The computed Hilbert kernel does not erase that support.

A compactly supported source realizes the full sharp phase loss

Fix $L > 0$, $0 \leq \eta < 1$, and the literal mass 7. Choose two nonzero smooth functions f_0, g_0 with disjoint compact supports in $(0, L)$. Write $\alpha = \int |f_0|^2 > 0$, $\beta = \int |g_0|^2 > 0$, and set

$$f = f_0/\sqrt{\alpha}, \quad g_1 = g_0/\sqrt{\beta}, \quad \psi_0(r) = \sqrt{7} f(r), \quad \psi_1(r) = \sqrt{7} \left(\eta f(r - L) + \sqrt{1 - \eta^2} g_1(r) \right). \quad (\text{HDI22})$$

Functions are extended by zero off their supports. The factors α, β explicitly describe the isometric coordinates of this constructed source; its Gram retains the mass 7. Disjoint supports give $\langle f, g_1 \rangle = 0$ and $\langle f, f(\cdot - L) \rangle = 0$, while $\|f\| = \|g_1\| = \|f(\cdot - L)\| = 1$. Direct integration therefore gives $M = 7I_2$ for $\Psi(v_0, v_1) = v_0 \psi_0 + v_1 \psi_1$. All exponential weighted Grams and derivative moments are finite, because these are smooth compactly supported columns.

On $0 \leq r < L$, only the $a = 0$ terms of f, g_1 and the $a = 1$ term of $f(\cdot - L)$ contribute to the actual holonomy transform $\sum_a e^{ia\theta} \psi(r + aL)$. Hence

$$\mathcal{Z}_{L,\theta} \psi_0 = \sqrt{7} f, \quad \mathcal{Z}_{L,\theta} \psi_1 = \sqrt{7} (\eta e^{i\theta} f + \sqrt{1 - \eta^2} g_1), \quad M_\theta = 7 \begin{pmatrix} 1 & \eta e^{i\theta} \\ \eta e^{-i\theta} & 1 \end{pmatrix}. \quad (\text{HDI23})$$

Its mean is exactly $7I_2$. The off-diagonal relative matrix squares to $\eta^2 I_2$, so its relative eigenvalues are $1 - \eta, 1 + \eta$. This is an entire trigonometric matrix family, obtained from the displayed source with the original sign convention for the phase.

Take $J = \begin{pmatrix} 1 & 0 \end{pmatrix}$, $B = \begin{pmatrix} 0 & 1 \end{pmatrix}^T$. Solving $JC_\theta = 1$ and $B^*M_\theta C_\theta = 0$, and integrating the original relation discrepancy, gives

$$\begin{aligned} C &= \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad C_\theta = \begin{pmatrix} 1 \\ -\eta e^{-i\theta} \end{pmatrix}, \quad X_\theta = \eta e^{-i\theta}, \\ G &= 7, \quad G_\theta = 7(1 - \eta^2), \quad X_\theta^* B^* M_\theta B X_\theta = 7\eta^2, \quad \mathcal{D} = 7\eta^2. \end{aligned} \tag{HDI24}$$

Each formula follows by matrix multiplication, and $C - C_\theta = BX_\theta$ is the actual relation identity. Thus equality in (H33) holds on the full phase circle in this class of smooth compactly supported Zak sources. The explicit arrow $\Psi : \mathbb{C}^2 \rightarrow L^2(\mathbb{R})$ and its quotient J specify this calibration's domain. They do not identify these columns with a particular arithmetic packet $F_h^{\otimes k}$. The arithmetic application continues to use its original $\mathcal{U}_k \mathcal{V}, J_N, \chi$ and the theta primitives of (H7), (H26), and (H53)–(H55).

Continuous common-image holonomy descent

Its contraction, metric gap, and completed receiving fibres

This continuation proves the continuous-phase version of H29–H31 and the exact maps to the completed targets in H64 of the delivered *All-holonomy descent of the original theta quotient*. The delivered NOTE.tex was read completely, all 1152 lines; its SHA-256 is

444f82a8fa420e6f471ef448442085f3a93acb137069e1505d155faab051b881.

The archive SHA-256 is

f0506a27cd4c9012f43fa70c3d1f8f996bbb6db01b0eea2944b34dcf290ca8f0.

All source coordinates, masses, phases, relation columns, and full primary multiplicities below are retained. No finite test or Lean execution is used as a proof. The sampled-density argument uses the full analytic proof *Periodized source density review*, PSD8–PSD14, whose TeX SHA-256 is

51102189bcc7a45610464143a5c0bee21c08059c317024e6e856bc7de993fe74.

The completed-fibre calculation develops H64 and is also proved in the companion completed-quotient and holonomy-intake derivations. Its repetition here supplies the maps needed for this continuous common-image complex, including the mean-zero contraction and the precise weighted relation gap.

Original coefficient spaces and actual phase norms

Fix the original monic polynomial χ , its full degree $q \geq 1$, and an integer $N \geq q - 1$. Put

$$\begin{aligned} E &= \mathbb{C}[S]/(\chi), & P_N &= \mathbb{C}[S]_{\leq N}, & J &= J_N : P_N \longrightarrow E, \\ B_N &= \ker J = \chi P_{N-q}, & A &= M_S : E \longrightarrow E, \end{aligned} \quad (\text{HCD1})$$

where $P_{-1} = 0$. Use the original increasing-power coefficient bases. Let $B : \mathbb{C}^r \rightarrow P_N$, $r = N - q + 1$, be multiplication by the literal χ , so that $\text{im } B = B_N$. At $r = 0$ it is the unique map from the zero coefficient space.

Let $\Theta = [0, 2\pi)$ carry its actual Fourier phase measure $d\mu(\theta) = d\theta/(2\pi)$. Retain the measurable Hermitian phase source matrices M_θ and their mean

$$M = \int_{\Theta} M_\theta d\mu, \quad 0 < aI_{P_N} \preceq M_\theta \preceq bI_{P_N} < \infty \quad \text{almost everywhere,} \quad a, b > 0. \quad (\text{HCD2})$$

The original source isometry H13–H17 identifies M with the original source Gram, including its full constant mass. For the actual source, the bounds in (HCD2) are obtained from H39–H41, also proved in HG3, as follows. Let $\Psi_N : P_N \rightarrow L^2(\mathbb{R}; \mathcal{K})$ be the original finite source columns after the half-density map, with $\mathcal{K} = L^2(\mathbb{R}^{k-1}, dz)$. Fix $\beta > 0$ for the retained weighted source Grams, whose finite entries are supplied by the original source regularity:

$$M_{\beta, \pm} = (e^{\pm\beta r} \Psi_N)^* (e^{\pm\beta r} \Psi_N), \quad \kappa_{\beta, N} = \max_{v \neq 0} \frac{v^* (M_{\beta, +} + M_{\beta, -}) v}{v^* M v}. \quad (\text{HCD2a})$$

These are the original weights $x_k^{\pm 2\beta}$; all original source measures remain present. The maximum is finite and positive because $M > 0$ and the two weighted Grams are finite positive forms. For $\mathcal{C}(s) = \int \Psi_N(r)^* \Psi_N(r+s) dr$ and $s \geq 0$, write the integrand at v as the inner product of $e^{-\beta r} \Psi_N(r)v$ and $e^{\beta(r+s)} \Psi_N(r+s)v$, multiplied by $e^{-\beta s}$. Cauchy–Schwarz, followed by translation in the second integral, gives

$$|v^* \mathcal{C}(s) v| \leq e^{-\beta s} \sqrt{(v^* M_{\beta, -} v)(v^* M_{\beta, +} v)}.$$

For $-s$ the same bound follows from $\mathcal{C}(-s) = \mathcal{C}(s)^*$. Unfolding the literal Zak sum gives $M_\theta = \sum_{j \in \mathbb{Z}} e^{ij\theta} \mathcal{C}(jL)$. The correlation bound and polarization give absolute convergence of each matrix

entry, uniformly in phase. Pairing j and $-j$, using $2\sqrt{uv} \leq u + v$, and summing the geometric series prove

$$-\delta_L M \preceq M_\theta - M \preceq \delta_L M, \quad \delta_L = \frac{\kappa_{\beta,N}}{e^{\beta L} - 1}. \quad (\text{HCD2b})$$

For any chosen $0 < \eta < 1$, the explicit finite circumference

$$L \geq \frac{1}{\beta} \log(1 + \kappa_{\beta,N}/\eta) \implies \delta_L \leq \eta < 1 \quad (\text{HCD2c})$$

therefore supplies (HCD2), with $a = (1 - \delta_L)\lambda_{\min}(M)$ and $b = (1 + \delta_L)\lambda_{\max}(M)$ in the original coefficient frame. The phase mean is still exactly M : every nonzero Fourier mode integrates to zero against $d\theta/(2\pi)$. The concrete choice $\eta = 1/2$ gives $L = \beta^{-1} \log(1 + 2\kappa_{\beta,N})$ and $\delta_L = 1/2$. These are actual finite-source bounds at the stated cutoff; no bound uniform in N or the arithmetic packet is asserted. The argument below also applies to any fixed L for which (HCD2) is already established.

Define the Hilbert space

$$\mathcal{P} = L^2(\Theta; P_N, M_\theta d\mu), \quad \|P\|_{\mathcal{P}}^2 = \int_{\Theta} P(\theta)^* M_\theta P(\theta) d\mu.$$

By (HCD2), its elements are exactly coefficientwise L^2 functions, with equivalent norms. Equality means equality almost everywhere. The coefficientwise Bochner integral $\text{av } P = \int P d\mu$ exists, since coefficientwise L^2 implies L^1 on this finite phase measure. Every fixed coefficient map commutes with this integral, entry by entry.

The closed relation space and common-image space are

$$\mathcal{B} = L^2(\Theta; B_N), \quad \mathcal{P}^{\text{eq}} = \{P \in \mathcal{P} : JP(\theta) = x \text{ a.e. for a fixed } x \in E\}. \quad (\text{HCD3})$$

They carry the restricted source norms. They are closed: J is continuous on coefficientwise L^2 , $B_N = \ker J$ is closed, and constant E -valued functions form a closed subspace. For the last assertion, averaging followed by repetition is a bounded idempotent with precisely that image.

The continuous cochain splitting and its full contraction

Use complexes in degrees 0, 1 with actual inclusion differentials:

$$\mathcal{C}_N = [B_N \hookrightarrow P_N], \quad \mathcal{C}_N = [\mathcal{B} \hookrightarrow \mathcal{P}^{\text{eq}}]. \quad (\text{HCD4})$$

Repetition $\Delta v(\theta) = v$ and averaging av are cochain maps. In degree one, $J \text{av } P = \int JP d\mu = x$; in degree zero the mean belongs to B_N . The equality $\text{av } \Delta = I$ uses the literal phase measure. Repetition is an isometry:

$$\|\Delta v\|_{\mathcal{P}}^2 = v^* \left(\int M_\theta d\mu \right) v = v^* M v.$$

Averaging is bounded with the explicit estimate

$$\|\text{av } P\|_M^2 \leq b \|\text{av } P\|_{\text{coeff}}^2 \leq b \int \|P(\theta)\|_{\text{coeff}}^2 d\mu \leq (b/a) \|P\|_{\mathcal{P}}^2. \quad (\text{HCD5})$$

This uses scalar Cauchy–Schwarz for the coefficient vector and does not replace the arithmetic mass by one.

Put $\mathcal{B}^0 = \{b \in \mathcal{B} : \text{av } b = 0\}$. The inverse continuous coefficient maps of complexes are

$$\begin{aligned} \mathcal{C}_N &\simeq \mathcal{C}_N \oplus [\mathcal{B}^0 \xrightarrow{I} \mathcal{B}^0], \\ P &\longmapsto (\text{av } P, P - \Delta \text{av } P), \quad (v, b^0) \longmapsto \Delta v + b^0. \end{aligned} \quad (\text{HCD6})$$

In degree one, $J(P - \Delta \text{av } P) = x - x = 0$ and its mean is zero. In degree zero every value already belongs to B_N . Both composites restore the original vector. Inclusion carries these components to inclusion and identity, respectively, proving the isomorphism of complexes.

The degree-minus-one map

$$h : \mathcal{P}^{\text{eq}} \longrightarrow \mathcal{B}, \quad hP = P - \Delta \text{av } P \quad (\text{HCD7})$$

satisfies $dh + hd = I - \Delta \text{av}$. In degree one this is $dhP = P - \Delta \text{av } P$; in degree zero it is $hdb = b - \Delta \text{av } b$. On the complementary complex it is the identity from degree one to degree zero. Boundedness follows from (HCD5) and the isometry of repetition. It is not asserted orthogonal or isometric.

The common quotient map is

$$j : \mathcal{P}^{\text{eq}} \longrightarrow E, \quad P \longmapsto JP(\theta) \text{ a.e.} \quad (\text{HCD8})$$

It is onto by repeating a right inverse of J , and its kernel is exactly $\mathcal{B}$. Inclusion is injective. Therefore $H^0(\mathcal{C}_N) = 0$ and $H^1(\mathcal{C}_N) = E$, with the isomorphism induced by the original full J . Repetition and averaging induce the identity on this E .

Multiplication by S maps P_N to P_{N+1} and B_N to B_{N+1} . Applied pointwise, it sends common image x to common image Ax and commutes with repetition, mean and h . These are maps between the stated cutoffs. On degree-one cohomology the action is the full companion A , including its nilpotent primary parts. No trace of the infinite-dimensional contractible complement is assumed.

The minimum norm of the common image

Retain the original and phasewise minimum sections

$$\begin{aligned} G &= (JM^{-1}J^*)^{-1}, \quad C = M^{-1}J^*G, \\ G_\theta &= (JM_\theta^{-1}J^*)^{-1}, \quad C_\theta = M_\theta^{-1}J^*G_\theta, \quad \overline{G} = \int G_\theta d\mu. \end{aligned} \quad (\text{HCD9})$$

Surjectivity and positivity make every inverse defined. Continuity of matrix inversion proves measurability. For any fixed coefficient right inverse C_{ref} of J ,

$$\frac{a}{\|J\|_{\text{coeff}}^2} I_E \preceq G_\theta \preceq b \|C_{\text{ref}}\|_{\text{coeff}}^2 I_E. \quad (\text{HCD10})$$

Indeed any lift p of x has $\|x\| \leq \|J\|\|p\|$, proving the lower bound, and the particular lift $C_{\text{ref}}x$ proves the upper bound. The formula for C_θ then also gives boundedness. All following integrals and sections are consequently defined.

Direct multiplication gives $JC_\theta = I$, and for $b_0 \in B_N$,

$$b_0^* M_\theta C_\theta = b_0^* J^* G_\theta = 0. \quad (\text{HCD11})$$

The section $\mathcal{S}x(\theta) = C_\theta x$ has common image x . Any other P with that image has $P - \mathcal{S}x \in \mathcal{B}$. Phasewise orthogonality gives

$$\|P\|_{\mathcal{P}}^2 = x^* \overline{G} x + \|P - \mathcal{S}x\|_{\mathcal{P}}^2. \quad (\text{HCD12})$$

Thus the quotient metric of the continuous complex is exactly $\overline{G} > 0$ on its full E . For the constant-source complex the identical argument gives G . Repetition is isometric on sources, and its induced identity $(E, G) \rightarrow (E, \overline{G})$ has the additional continuous relation directions available in the target.

The positive relation gap and its mean-zero decomposition

For $r > 0$, put $X_\theta = (B^* M_\theta B)^{-1} B^* M_\theta C$. Both sections have right-inverse image I_E , so their difference lies in B_N and equals BX_θ ; multiplication by $B^* M_\theta$ proves the formula and uniqueness.

At $r = 0$, both sections equal the unique inverse of J and X_θ is the zero map into the zero space. Expanding $C = C_\theta + BX_\theta$ and using (HCD11) proves

$$\boxed{G - \overline{G} = \mathcal{D} = \int (C - C_\theta)^* M_\theta (C - C_\theta) d\mu} \quad (\text{HCD13})$$

$$= \int X_\theta^* (B^* M_\theta B) X_\theta d\mu \succeq 0.$$

The integral of $C^* M_\theta C$ is $C^* M C = G$. Every term is the norm of a field in the original relation image. Multiplication by the literal χ , followed by the inherited finite tensor theta primitive, realizes that field in the original cochain complex.

The relation to the particular mean-zero contraction is also exact. Define

$$\overline{C} = \int C_\theta d\mu, \quad a_C = C - \overline{C} : E \rightarrow B_N, \quad h_\theta = C_\theta - \overline{C} : E \rightarrow B_N. \quad (\text{HCD14})$$

Then $\int h_\theta d\mu = 0$, and h_θ is precisely the field h_θ . The original discrepancy is $C - C_\theta = a_C - h_\theta$ and need not have mean zero. Since a_C has relation-valued columns, phase and common-source orthogonality give

$$\begin{aligned} \int a_C^* M_\theta h_\theta d\mu &= -a_C^* M \overline{C} \\ &= -a_C^* M (C - a_C) = a_C^* M a_C. \end{aligned} \quad (\text{HCD15})$$

The adjoint cross term is the same Hermitian matrix. Expanding all terms of (HCD13) therefore proves

$$\boxed{\mathcal{D} = \int h_\theta^* M_\theta h_\theta d\mu - a_C^* M a_C.} \quad (\text{HCD16})$$

This retains the positive relation norm and the complete weighted cross terms. A mean-zero coefficient field is not presumed orthogonal to constant relations for the varying metric. At $N = q - 1$, $B_N = 0$, $C_\theta = C = J^{-1}$, $a_C = h_\theta = 0$ and $\overline{G} = G$.

Actual sampled Hilbert spaces, including zero weights

Fix the original circumference $L > 0$, tensor degree $k \geq 1$, and $c = k/2$. Retain

$$\begin{aligned} u_n(\theta) &= (2\pi n + \theta)/L, & s_n(\theta) &= c + iu_n(\theta), \\ w_n(\theta) &= \frac{2\pi}{L} m_{h,k}(u_n(\theta)) \geq 0, & n &\in \mathbb{Z}. \end{aligned} \quad (\text{HCD17})$$

The Fourier coefficient is literally $2\pi/L$, with $m_{h,k} = w_h^k$, $w_h = |(g/h)(1/2 + iu)|^2/(2\pi)$ and $g = 2\xi$. Sampling gives

$$\mathsf{T}_\theta P = (P(s_n(\theta)))_n, \quad \|\mathsf{T}_\theta P\|_\theta^2 = \sum_n w_n(\theta) |P(s_n(\theta))|^2 = P^* M_\theta P. \quad (\text{HCD18})$$

This is the actual phase source Gram H17.

Define $\mathcal{H}_\theta$ to be weighted square-summable sequences modulo sequences supported where $w_n(\theta) = 0$. Thus zero weights, possible when $k = 1$, have an explicit null subspace. For $k \geq 2$, positivity of the original convolution makes every sample weight positive. At $k = 1$, retain the exact ideal equality $(\chi_{h,1}) = (h)$. If $a_h \neq 0$ is the original leading coefficient of h , then $\chi_{h,1} = h/a_h$ is its monic generator, and $\chi_{h,1} = h$ in the original monic packet convention. Indeed a polynomial annihilates multiplication by S on $\mathbb{C}[S]/(h)$ if and only if its action on $[1]$ is zero, which is equivalent to belonging to (h) . The analytic source remains the original g/h , with the factor a_h retained in that quotient and its norm. Now take a complete selected root $\rho = 1/2 + iu_n(\theta)$ whose common order in

g and h is m . The two Taylor expansions have leading coefficients $g^{(m)}(\rho)/m!$ and $h^{(m)}(\rho)/m!$, so the canceled entire quotient has the exact nonzero value

$$(g/h)(\rho) = \frac{g^{(m)}(\rho)}{h^{(m)}(\rho)}, \quad w_n(\theta) = \frac{1}{L} \left| \frac{g^{(m)}(\rho)}{h^{(m)}(\rho)} \right|^2 > 0. \quad (\text{HCD18a})$$

This retains both cancellation factorials and the full Fourier mass $(2\pi/L)(1/(2\pi))$. It is the selected-root calculation PSD5 and PSD41. Other $k = 1$ coordinates where g/h vanishes still give the explicit null subspace above. For a concrete measurable realization use the isometry $v \mapsto (\sqrt{w_n(\theta)}v_n)_n$ into ordinary $\ell^2(\mathbb{Z})$, with zero coordinates at zero weights. Its inverse on its range is $v_n = y_n/\sqrt{w_n(\theta)}$ at positive weights; other coordinates give the null class. This specifies the norm map without changing a mass. The range projections are measurable diagonal matrices; their images of the countable standard basis give a measurable field.

The original gamma-contour and alternating-zeta calculation PSD8–PSD13 proves, for every $0 < \gamma < \pi/2$, the finite constants

$$A_\gamma = \|e^{\gamma|\cdot|}w_h\|_\infty, \quad B_\gamma = \|e^{\gamma|\cdot|}w_h\|_1.$$

That calculation retains the leading coefficient of h , its full degree, and the continuous canceled values on the compact interval containing its selected roots. With $f_\gamma(v) = e^{\gamma|v|}w_h(v)$, the ordinary triangle inequality on the convolution variables and integration of the remaining factors give, for every real u ,

$$e^{\gamma|u|}m_{h,k}(u) \leq f_\gamma^{*k}(u) \leq A_\gamma B_\gamma^{k-1}. \quad (\text{HCD19a})$$

For the second inequality bound one factor by A_γ and integrate each of the other $k - 1$ factors with its literal Lebesgue measure. For $k = 1$ the expression is A_γ , with $B_\gamma^0 = 1$. For any $0 < 2\epsilon < \gamma$,

$$\sup_{0 \leq \theta < 2\pi} \sum_n w_n(\theta) e^{2\epsilon|u_n(\theta)|} < \infty. \quad (\text{HCD19})$$

Indeed $|u_n(\theta)| \geq (2\pi/L)(|n| - 1)$, so with $d = \gamma - 2\epsilon > 0$ the sum is at most

$$\frac{2\pi}{L} A_\gamma B_\gamma^{k-1} e^{2\pi d/L} \left(1 + \frac{2e^{-2\pi d/L}}{1 - e^{-2\pi d/L}} \right).$$

For every fixed polynomial Q , the constant $\sup_{u \in \mathbb{R}} |Q(c + iu)|^2 e^{-d|u|/2}$ is finite. Inserting it into the same geometric estimate, with $d/2$ in place of d , proves uniform exponential summability of $w_n(\theta)|Q(s_n(\theta))|^2$. This includes $Q = \chi$ and every fixed polynomial relation used below. All original constants in the weights are unchanged. The full source proof of PSD8–PSD14 is a dependency of this continuation; the remaining discrete density argument is given explicitly here with its shifted Fourier phase.

Density and the complete fibrewise Hilbert quotient

Here is a direct density proof with the zero weights retained. If $f \in \mathcal{H}_\theta$ is orthogonal to all polynomials, put $a_n = w_n(\theta)\overline{f_n}$. Cauchy–Schwarz and (HCD19) give $\sum |a_n| e^{\epsilon|u_n|} < \infty$. Hence $F(z) = \sum_n a_n e^{zu_n}$ is holomorphic on $|\Re z| < \epsilon$, with termwise derivatives on smaller closed strips. Every derivative at zero is zero, since polynomials in S and in $(S - c)/i$ span the same space. The Taylor series and the identity theorem give $F = 0$ on the connected strip. On the imaginary axis,

$$F(it) = e^{it\theta/L} \sum_n a_n e^{2\pi int/L}.$$

The series is absolutely uniformly convergent. Multiplication by $e^{-it\theta/L} e^{-2\pi imt/L}$ and integration over $0 \leq t \leq L$ give $a_m = 0$. At positive weights this implies $f_m = 0$; at zero weights the class is

already null. Thus polynomials are dense. The identical argument proves polynomial density for the weights $w_n|\chi(s_n)|^2$.

Multiplication by the original polynomial is the isometry

$$U_\chi : \ell^2(w_n|\chi(s_n)|^2) \longrightarrow \mathcal{H}_\theta, \quad f \longmapsto (\chi(s_n)f_n)_n. \quad (\text{HCD20})$$

Its image is precisely the sequences vanishing at the positive-weight sampled roots of χ . To prove the converse inclusion, take such a sequence v and set $f_n = v_n/\chi(s_n)$ when $\chi(s_n) \neq 0$, and zero elsewhere. Its squared domain norm equals $\|v\|_\theta^2$. Thus the image is closed. Polynomial density in the domain now proves

$$\mathcal{R}_\theta := \overline{\chi\mathbb{C}[S]}^{\mathcal{H}_\theta} = \{v : v_n = 0 \text{ if } w_n(\theta) > 0, \chi(s_n(\theta)) = 0\}. \quad (\text{HCD21})$$

The receiving Hilbert quotient is exactly

$$\begin{aligned} \mathcal{Q}_\theta &= \mathcal{H}_\theta / \mathcal{R}_\theta \simeq \bigoplus_{\substack{n: \chi(s_n(\theta))=0 \\ w_n(\theta)>0}} \mathbb{C}, \\ \|v\|_{\mathcal{Q}_\theta}^2 &= \sum_{\text{displayed } n} w_n(\theta) |v_n|^2. \end{aligned} \quad (\text{HCD22})$$

The quotient map is orthogonal coordinate restriction. No weight equal to zero is inverted.

The complete primary kernel before phase integration

Let Λ_θ^+ be the distinct sampled roots of χ with positive weight, and set $p_\theta(S) = \prod_{\lambda \in \Lambda_\theta^+} (S - \lambda)$, using $p_\theta = 1$ for an empty product. The finite map is

$$\rho_\theta : E \longrightarrow \mathcal{Q}_\theta, \quad [P]_\chi \longmapsto (P(\lambda))_{\lambda \in \Lambda_\theta^+}. \quad (\text{HCD23})$$

It is well defined, and is onto by Lagrange interpolation at these distinct points, using degree less than $|\Lambda_\theta^+| \leq q$. Vanishing at all those points is equivalent to divisibility by p_θ . Since $p_\theta \mid \chi$,

$$\ker \rho_\theta = (p_\theta)/(\chi) \subset E. \quad (\text{HCD24})$$

This specifies the zero-weight-corrected kernel in H64: a root at a zero-mass coordinate contributes no receiving Hilbert line.

Write the full original factorization as $\chi = \prod_{\lambda \in \Lambda} (S - \lambda)^{m_\lambda}$. The complete Taylor map is

$$\begin{aligned} E &\longrightarrow \prod_{\lambda \in \Lambda} \mathbb{C}[\varepsilon_\lambda]/(\varepsilon_\lambda^{m_\lambda}), \\ [P] &\longmapsto \left(\sum_{j=0}^{m_\lambda-1} \frac{P^{(j)}(\lambda)}{j!} \varepsilon_\lambda^j \right)_\lambda. \end{aligned} \quad (\text{HCD25})$$

For an explicit inverse, let $q_\lambda = \chi/(S - \lambda)^{m_\lambda}$ and let u_λ be the Taylor polynomial of $1/q_\lambda$ at λ through degree $m_\lambda - 1$. Then $e_\lambda = [u_\lambda q_\lambda]_\chi$ is one in the λ factor and zero in every other factor, by divisibility by each original prime power. It follows that $\sum e_\lambda = 1$ and the factors are orthogonal idempotents. The inverse is $\sum_\lambda e_\lambda \sum_j a_{\lambda,j} (S - \lambda)^j$.

In these full primary coordinates, ρ_θ takes the constant coefficient of each active sampled factor. Its kernel in that factor is (ε_λ) , retaining every order $1, \dots, m_\lambda - 1$. For an unsampled factor or a zero-weight sampled factor, the kernel is the entire primary algebra, including its constant term. The original action is multiplication by $\lambda + \varepsilon_\lambda$; the receiving action is the simple scalar λ . Evaluation intertwines these actions with the displayed nilpotent kernel.

The measurable completed field and its vanishing map

Every sampled root has real part c . For a fixed root λ with $\Re \lambda = c$, its only possible phase is

$$\theta_\lambda = L\Im \lambda \pmod{2\pi}, \quad 0 \leq \theta_\lambda < 2\pi, \quad n_\lambda = (L\Im \lambda - \theta_\lambda)/(2\pi) \in \mathbb{Z}.$$

Thus the set Θ_χ of possible phases is finite. For $\theta \notin \Theta_\chi$, $\mathcal{Q}_\theta = 0$. At an exceptional phase the positive-weight condition in (HCD22) is still retained.

In the measurable realization after (HCD18), the projection onto $\mathcal{Q}_\theta$ has diagonal entries

$$\mathbf{1}_{\{w_n(\theta) > 0\}} \mathbf{1}_{\{\chi(s_n(\theta)) = 0\}}, \quad n \in \mathbb{Z}.$$

These are measurable and give an explicit field of projections. The direct integral satisfies

$$\mathcal{Q} = \int_{\Theta}^{\oplus} \mathcal{Q}_\theta d\mu = 0, \quad (\text{HCD26})$$

because every measurable section has squared norm supported on the finite measure-zero set Θ_χ . This is an almost-everywhere assertion; each exceptional nonzero receiving line is still exactly the weighted line in (HCD22).

For $x \in E$, the section $\theta \mapsto \rho_\theta x$ is measurable: it is given by polynomial evaluations and the displayed coordinate masks. It defines

$$\rho : E \longrightarrow \mathcal{Q}, \quad x \longmapsto [\theta \mapsto \rho_\theta x]_{\text{a.e.}}, \quad \ker \rho = E. \quad (\text{HCD27})$$

Every pointwise exceptional value has zero direct-integral class. This does not assert that those pointwise maps vanish, or that the original nonreduced coefficient algebra was zero.

The source-to-completion cochain map

The direct-integral source is the weighted space with squared norm $\int_{\Theta} \sum_n w_n(\theta) |v_n(\theta)|^2 d\mu$. Let $\mathcal{H}$ denote this space, and let $\mathcal{R}$ be its closed subspace of sections belonging to $\mathcal{R}_\theta$ almost everywhere. The coordinate projection above has kernel $\mathcal{R}$.

The measurable evaluation map is

$$\mathsf{T} : \mathcal{P} \longrightarrow \mathcal{H}, \quad P \longmapsto (P(\theta, s_n(\theta)))_{\theta, n}.$$

The integrated identity (HCD18) proves it is an isometry. Evaluation of a relation-valued section is in $\mathcal{R}$, because the original section is a multiple of the literal χ . Hence there is a cochain map

$$[\mathcal{B} \hookrightarrow \mathcal{P}^{\text{eq}}] \longrightarrow [\mathcal{R} \hookrightarrow \mathcal{H}], \quad (\text{HCD28})$$

using evaluation in both degrees. On a common image x , the receiving quotient value is $\rho_\theta x$, since representatives differ by an original relation. The induced degree-one map is exactly (HCD27).

The global relation subspace is also the closure of measurable finite polynomial relation sections. To prove this without a measurable choice of approximating polynomials, suppose $v \in \mathcal{H}$ is orthogonal to every section $\mathbf{1}_U(\theta) \chi(S) S^j$, for measurable U and $j \geq 0$. Its fibre inner product with χS^j is integrable by Cauchy–Schwarz, and its integral over every U is zero, so that inner product is zero almost everywhere. The original exponential sample bounds make every fixed polynomial section square-integrable. Intersecting the countably many resulting full-measure sets, (HCD21) gives $v(\theta) \in \mathcal{Q}_\theta$ there. Conversely every such section is orthogonal to the relations. The orthogonal-complement characterization proves that their closed span is exactly $\mathcal{R}$.

Since $\mathcal{Q} = 0$, $\mathcal{R} = \mathcal{H}$ as Hilbert spaces, and the completed complex is the identity complex. Its degree-one cohomology is zero. The finite continuous common-image complex has positive Gram $\overline{G}$ on its full E by (HCD8)–(HCD12). The intervening map and its whole kernel are precisely (HCD23)–(HCD28).

The receiving supported zero and all marked fibres

Integrate in the stated coefficient fibre, then apply the original marked support construction. At a fixed original label ℓ , a linear map f has the lift

$$(\ell, x) \longmapsto (\ell, f(x)), \quad \tau \longmapsto \tau. \quad (\text{HCD29})$$

For (HCD27), every represented input maps to the receiving supported zero $(\ell, 0)$, and external absence maps to external absence τ . Their inverse images are respectively the full represented fibre and the singleton external absence. At a fixed phase, the represented kernel is the entire ideal (HCD24), with its full primary decomposition (HCD25). No integral on arbitrary mixed-support totals is presumed.

If the empty-packet case $q = 0$ is separately admitted, $E = 0$, $B_N = P_N$, and the finite and continuous coefficient complexes are identity complexes. The quotient Gram determinant is the empty determinant one. The analytic source and its literal norm remain present before the quotient. Its marked receiving object still has τ and the represented zero, with the same maps (HCD29).

Exact outcome and scope

The continuous common-image complex has the explicit bounded cochain contraction (HCD7) on its complement and retains the whole nonreduced quotient in degree one. Its minimum metric is $\overline{G}$, and its difference from the constant-source metric is the positive original relation norm (HCD13). The mean-zero decomposition retains the weighted cross term (HCD15), giving (HCD16).

The subsequent completion has receiving weighted value spaces (HCD22), primary kernels (HCD24)–(HCD25), measurable field (HCD26), and actual cochain map (HCD28). Its receiving coefficient direct integral is zero because only finitely many phases can sample a fixed polynomial root. Every exceptional phase, nilpotent component and zero-weight coordinate remains in the stated maps and kernels. No arithmetic volume asymptotic or RH conclusion is inferred.

Original continuous source: completed relations and canonical degree limit

This continues the fixed-packet calculation HD1–9 without changing its measure, polynomial coordinate, quotient, multiplicities or metric. It supplies the continuous-source counterpart of the already proved sampled quotient P53–58 and H64. Every limit below fixes the original packet h and tensor degree k ; no tensor-degree rate or Riemann-hypothesis conclusion is asserted.

Inputs and exact maps

Let $g = 2\xi$, $k \geq 1$, $c = k/2$, and retain

$$w_h(t) = \frac{|(g/h)(1/2 + it)|^2}{2\pi}, \quad m(u) = w_h^{*k}(u), \quad d\nu(u) = m(u) du, \quad S(u) = c + iu. \quad (\text{CR1})$$

HD1–4 prove for this actual measure that, for any fixed $0 < b < \pi/2$, $\int e^{b|u|} d\nu(u) < \infty$. The nonzero density is positive almost everywhere, and ν has no atoms. Its total mass remains μ_h^k . Fix the original nonzero monic cyclic polynomial $\chi = \chi_{h,k}$, with degree $q \geq 1$ and every primary multiplicity retained. Define

$$H = L^2(\mathbb{R}, d\nu), \quad H_\chi = L^2(\mathbb{R}, |\chi(S(u))|^2 d\nu), \quad E = \mathbb{C}[S]/(\chi), \quad J : \mathbb{C}[S] \longrightarrow E. \quad (\text{CR2})$$

For $N \geq q - 1$, let $\mathcal{P}_N = \mathbb{C}[S]_{\leq N}$, $\mathcal{B}_N = \chi \mathcal{P}_{N-q}$, with $\mathcal{P}_{-1} = 0$. The source maps, their kernels and their arithmetic class maps are exactly

$$0 \longrightarrow \mathcal{P}_{N-q} \xrightarrow{M_\chi} \mathcal{P}_N \xrightarrow{J_N} E \longrightarrow 0. \quad (\text{CR3})$$

No coefficient of χ or of the polynomial source has been rescaled. The original source norm is the norm of H through the specified Mellin observation. In particular all conclusions transport back through that original isometry, rather than selecting another Hilbert norm.

Polynomial density, with its measure proof

We first prove the exact density fact used twice below. Let η be any of the two finite measures ν or $|\chi(S)|^2 \nu$. Both have a positive exponential moment. For the second, polynomial growth gives $|\chi(c + iu)|^2 \leq C(1 + |u|)^{2q}$; choosing $0 < a < b$ absorbs this factor into $e^{(b-a)|u|}$, so $\int e^{a|u|} d\eta < \infty$. This bound is a consequence for the fixed original measure, not a new source hypothesis. All polynomial moments consequently exist.

If f in $L^2(\eta)$ is orthogonal to all polynomials in S , it is orthogonal to all polynomials in u because $u = (S - c)/i$ is the exact inverse coordinate. Set $d\sigma = \bar{f} d\eta$. Cauchy–Schwarz gives

$$\int e^{a|u|/2} d|\sigma|(u) \leq \|f\|_{L^2(\eta)} \left(\int e^{a|u|} d\eta(u) \right)^{1/2} < \infty. \quad (\text{CR4})$$

Thus $F(z) = \int e^{izu} d\sigma(u)$ is holomorphic for $|\text{Im } z| < a/2$. On each smaller closed strip, powers of $|u|$ are absorbed by the remaining exponential margin, justifying every derivative under the integral. All derivatives at zero vanish by the orthogonality equations. Its Taylor series and the identity theorem give $F = 0$ on the connected strip, including the real axis.

Here is the required uniqueness step explicitly. For $t > 0$ put $\gamma_t(x) = (4\pi t)^{-1/2} e^{-x^2/(4t)}$. Its Fourier integral is $\gamma_t(x) = (2\pi)^{-1} \int e^{-t\zeta^2} e^{-i\zeta x} d\zeta$. Fubini, justified by the finite variation of σ and the integrable Gaussian factor, gives $\sigma * \gamma_t = 0$. For every continuous compactly supported ϕ , $\phi * \gamma_t \rightarrow \phi$ uniformly: uniform continuity controls $|y| < \epsilon$, and the Gaussian tail controls its complement. Integrating against σ gives $\int \phi d\sigma = 0$. Uniqueness of finite regular measures on

continuous compactly supported test functions gives $\sigma = 0$. Hence $f = 0$ η -almost everywhere. The orthogonal complement of the polynomial span is zero, proving

$$\overline{\mathbb{C}[S]}^H = H, \quad \overline{\mathbb{C}[S]}^{H_\chi} = H_\chi. \quad (\text{CR5})$$

The original relation multiplication is an onto isometry

The map and its inverse on equivalence classes are

$$U_\chi : H_\chi \longrightarrow H, \quad f \longmapsto \chi(S)f, \quad U_\chi^{-1}v(u) = \frac{v(u)}{\chi(S(u))} \quad (\chi(S(u)) \neq 0). \quad (\text{CR6})$$

At the finitely many real u where $\chi(S(u)) = 0$, choose the inverse value zero. These points have ν -measure zero, so the choice does not affect either Hilbert equivalence class. Direct integration proves

$$\|U_\chi f\|_H^2 = \int |f|^2 |\chi(S)|^2 d\nu = \|f\|_{H_\chi}^2, \quad \|U_\chi^{-1}v\|_{H_\chi}^2 = \|v\|_H^2. \quad (\text{CR7})$$

The two composites are identities almost everywhere. Polynomial density in H_χ and this actual multiplier therefore give

$$\boxed{\overline{\chi\mathbb{C}[S]}^H = H, \quad H/\overline{\chi\mathbb{C}[S]} = 0.} \quad (\text{CR8})$$

The coordinate-level comparison from the original algebraic quotient is $[P]_\chi \mapsto P + \overline{\chi\mathbb{C}[S]} = 0$, with kernel all of E . This identifies the map, not the two quotients. More precisely, complete both terms of $[\chi\mathbb{C}[S] \hookrightarrow \mathbb{C}[S]]$ in their inherited H norms. Both completions identify with H by CR5/CR8, and the extended inclusion is I_H . The explicit degree-minus-one contraction of $[H \xrightarrow{I_H} H]$ is I_H . Before completion, the cohomology is the full algebraic E ; its induced map to this completed cohomology is zero with the stated kernel.

Limit of the same finite canonical metric

Use the fixed increasing-power remainder section $s : E \rightarrow \mathcal{P}_{q-1}$, the inverse of J on that finite subspace. Let Π_N be the H -orthogonal projection onto the finite original relation subspace $\mathcal{B}_N$. The exact least-norm lift and quotient Gram are

$$R_N = (I - \Pi_N)s, \quad J_N R_N = I_E, \quad G_N = R_N^* R_N. \quad (\text{CR9})$$

To verify minimality directly, every lift of x is $s(x) + b$ with b in $\mathcal{B}_N$. The decomposition into $\mathcal{B}_N$ and its orthogonal complement shows that the unique minimizing lift is $R_N x$. The adjoints in the displayed Gram use the fixed original coefficient frame on E and the actual source H norm. If M_N is the original source Gram, the equivalent coordinate formulas are

$$G_N = (J_N M_N^{-1} J_N^*)^{-1}, \quad R_N = M_N^{-1} J_N^* G_N. \quad (\text{CR10})$$

Their equality with CR9 follows from $J_N R_N = I$ and source orthogonality to $\ker J_N$; the minimizer is unique. Every finite G_N is positive definite.

The nested $\mathcal{B}_N$ have dense union by CR8. For any v in H and $\epsilon > 0$, choose b in some $\mathcal{B}_n$ with $\|v - b\| < \epsilon$. For $N \geq n$, $\|(I - \Pi_N)v\| \leq \|v - b\| < \epsilon$. Thus Π_N converges strongly to I_H . Applying this to each of the q fixed columns of s proves

$$\boxed{G_N \downarrow 0, \quad V_N = \det G_N \longrightarrow 0 \quad (N \longrightarrow \infty; \ h, k \text{ fixed}).} \quad (\text{CR11})$$

The decrease is Loewner decrease, because enlarging the admitted relation space cannot increase a minimum. Convergence is in every fixed finite coordinate matrix norm: each column norm of

R_N tends to zero, and $|(G_N)_{ij}| \leq \|R_N e_i\| \|R_N e_j\|$. Determinant convergence then follows from its polynomial expression. This proves no growth rate.

For each nonzero x in E , the literal sequence $R_N x$ tends to zero in H but has $J(R_N x) = x$ for every N . Therefore the densely defined map $J : \mathbb{C}[S] \subset H \rightarrow E$, with any fixed positive finite norm on its original coefficient target, is not closable. Every $(0, x)$ lies in the closure of its graph. This is stronger than one evaluation witness, and records exactly why the algebraic class map has no continuous extension to this completed Hilbert quotient. It does not make J zero on its original polynomial domain.

In fact the graph closure is all of H times E . Given (v, y) , choose polynomials p_j tending to v by CR5, and put $x_j = y - Jp_j$. For each fixed j , CR9–11 allow a cutoff N_j with $\|R_{N_j} x_j\|_H < 1/j$. Then the polynomial $p_j + R_{N_j} x_j$ tends to v and its exact J value is y . This proves the full graph assertion without a bound uniform in j or an extension of J to H .

Under any fixed coefficient change $x = Ux'$, the matrices and source lifts transform by $G'_N = U^* G_N U$, $R'_N = R_N U$, and the identity on classes transforms by $J'_N = U^{-1} J_N$. Hence CR9–11 retain precisely the same morphisms in another coordinate frame; no N -dependent frame or metric normalization is used.

The phase quotient and averaging comparison

Fix the original circumference L . No assertion that every $k = 1$ phase source Gram is invertible is needed. Write T_θ for the original polynomial evaluation into its positive-weight sampled Hilbert space H_θ . The precise finite receiving space and comparison map are

$$Q_{N,\theta} = T_\theta(\mathcal{P}_N)/T_\theta(\mathcal{B}_N), \quad \rho_{N,\theta} : E \longrightarrow Q_{N,\theta}, \quad [P]_\chi \longmapsto [T_\theta P], \quad \ker \rho_{N,\theta} = J_N(\ker(T_\theta|_{\mathcal{P}_N})). \quad (\text{CR12a})$$

Both images are finite-dimensional closed subspaces of H_θ and carry its actual norm. The map is onto. It is well defined because any two degree- N lifts differ by $\mathcal{B}_N$. If $T_\theta P$ lies in $T_\theta \mathcal{B}_N$, subtract the corresponding b in $\mathcal{B}_N$ to obtain an element of $\ker T_\theta$ with the same J_N class; this proves the displayed kernel in both directions. Its pulled-back squared norm is the uniquely specified positive semidefinite coefficient form

$$x^* G_{N,\theta} x = \inf_{b \in \mathcal{B}_N} \|T_\theta(sx + b)\|_{H_\theta}^2. \quad (\text{CR12b})$$

This is the same canonical quotient form: when the phase source Gram is positive, $\rho_{N,\theta}$ is an isomorphism and CR12b gives exactly the positive Gram $(J_N M_{N,\theta}^{-1} J_N^*)^{-1}$. When it is singular, CR12a states the actual additional kernel and CR12b uses no inverse. No zero-weight coordinate is inverted and no new metric is selected.

The form is measurable in θ : for each x its infimum may be taken over a countable dense set of relation coefficients, since its finite quadratic norm is continuous in those coefficients. Each such value is measurable from the original sampled measure. Polarization gives the matrix entries. Using the particular lift $R_N x$ in CR12b and the original all-phase source identity proves

$$\int x^* G_{N,\theta} x \frac{d\theta}{2\pi} \leq \int \|T_\theta R_N x\|_{H_\theta}^2 \frac{d\theta}{2\pi} = \|R_N x\|_H^2 = x^* G_N x. \quad (\text{CR12c})$$

In particular these integrals are finite. Define the actual mean form $\bar{G}_N = \int G_{N,\theta} d\theta / (2\pi)$ and its difference $D_N = G_N - \bar{G}_N$. Then, for all original $k \geq 1$,

$$G_N = \bar{G}_N + D_N, \quad D_N \succeq 0, \quad 0 \preceq \bar{G}_N \preceq G_N. \quad (\text{CR12})$$

On an admitted positive phase family, D_N agrees with the complete relation-gap formula HCD13, including its phasewise least-norm sections; CR12a–c extend the comparison through the specified kernels without asserting those inverse formulas at singular phases.

Consequently both $\bar{G}_N \rightarrow 0$ and $D_N \rightarrow 0$ in fixed coefficient norm, without interchanging any infinite-dimensional inverse and integral. The relative matrix $G_N^{-1}D_N$ is not assigned a zero limit by this assertion: only the preceding absolute inequalities have been proved. The actual relative transfer remains the previously calculated metric/generator commutator, not this unscaled matrix limit.

The independent sampled density argument P53–58/CQ14–22 gives, at each fixed phase, the exact limit

$$G_{N,\theta} \downarrow K_\theta = J_{Z_\theta}^* \text{diag}(\nu_{n,\theta}) J_{Z_\theta}, \quad Z_\theta = \{n : \nu_{n,\theta} > 0, \chi(S_{n,\theta}) = 0\}, \quad (\text{CR13})$$

where $S_{n,\theta} = c + i(2\pi n + \theta)/L$ and $\nu_{n,\theta} = (2\pi/L)m((2\pi n + \theta)/L)$. Its exact kernel is $(p_\theta)/(\chi)$, with p_θ the squarefree product of the displayed sampled roots. Both full primary multiplicities and all masses remain in the domain/map. The target takes only the stated scalar values.

If Z_θ is nonempty, choose an original class x whose evaluation at one of those roots is one (the class of 1 already suffices). Its canonical single-phase/source relative cost satisfies the explicit inequality

$$\frac{x^* G_{N,\theta} x}{x^* G_N x} \geq \frac{\nu_{n,\theta}}{x^* G_N x} \rightarrow \infty. \quad (\text{CR14})$$

This strengthens HD’s full-source evaluation bound to this precise quotient comparison. It needs an actual positive-mass sampled root; none is invented. For $k = 1$ full-order cancellation supplies positivity at any selected sampled root; for $k \geq 2$ the original convolution is positive everywhere. If Z_θ is empty, CR13 has zero limit but CR14 makes no relative-rate assertion.

For the fixed χ , the exceptional phases are finite. The completed phase quotient field is therefore zero almost everywhere for its actual phase measure, and its direct integral is zero. The exact source isometry over all phases, the finite quotient identity $J_N R_N = I$, CR11, CR12 and the exceptional positive quotient limit CR13 are all compatible through their displayed maps. No arithmetic class is declared absent before the specified completion. A retained split label is transported as $(\lambda, v) \mapsto (\lambda, fv)$; the zero comparison takes it to $(\lambda, 0)$, fixes τ , and does not identify that receiving zero with external absence.

Provenance and scope

Read in full before deriving this note: the owner HD1–9 source, SHA256 56c05d9be82fa356f82d71324f317f86325e761531767b57d4a085e91b7760fa; holonomy_completed_quotient_density_review_20260913.md; and holonomy_continuous_descent_review_20260913.tex. P53–58 and surrounding original-source definitions were read directly in their delivered research note. CR4–11 are proved here, not inferred from a numerical check. CR12–14 explicitly compose those results with the named existing phase maps.

This is a written argument, pending independent review. No new Lean, numerical arithmetic enclosure, publication, tensor- k estimate or claimed solution of the remaining arithmetic upper-bound problem accompanies it.